Probabilistic Graphical Models and Causal Inference

Instructor: Prof. M. Arashi

Contents

Session 1: Foundations: Conditional Independence 4

1 Why the Joint Distribution Alone Is Not Enough 4

2 Independence and Conditional Independence 4
2.1 Three Structural Properties 6

3 Worked Examples 6

Session 2: Bayesian Networks: DAGs and Independence Encoding 10

4 From Chain Rule to Graph 10

5 Reading Independence off the Graph: d-Separation 11
5.1 Three Building Blocks 11

6 Worked Examples 12

Session 3: Inference: Variable Elimination 16

7 The Inference Problem 16

8 Factors: The Currency of Computation 16
8.1 The Variable Elimination Algorithm 17

9 Elimination Order Matters: Treewidth 17

10 Worked Examples 18

Session 4: Markov Networks: Undirected Models 22

11 Gibbs Distributions 22

12 Reading Independence from an Undirected Graph 22

13 Bayesian Networks and Markov Networks Are Incomparable 23

14 Worked Examples 24

Session 5: Learning: Parameter Estimation and Latent Variables 28

15 Maximum Likelihood Estimation with Full Observability **28**

16 When Observability Fails: Latent Variables **29**

17 The EM Algorithm **30**

18 Worked Examples **30**

Session 6: Structure Learning and Approximate Inference **34**

19 Learning Structure: Why It's Hard, and What's Done in Practice **34**

20 When Exact Inference Fails: The Case for Sampling **35**
20.1 Forward (Ancestral) Sampling 35
20.2 Markov Chain Monte Carlo: Gibbs Sampling 35

21 Worked Examples **36**

Session 7: Correlation vs. Causation: The Conceptual Pivot **41**

22 Two Real Reversals **41**

23 The Graphical Diagnosis **42**

24 Why $P(Y \mid X{=}x)$ Is the Wrong Quantity **43**

25 Worked Examples **44**

Session 8: Structural Causal Models **47**

26 Structural Causal Models **47**

27 Intervention as Graph Surgery **48**

28 Counterfactuals: A Glimpse **49**

29 Worked Examples **50**

Session 9: Intervention and Identification: Do-Calculus **54**

30 The Identification Problem **54**

31 The Backdoor Criterion **54**

32 The Front-Door Criterion: When Adjustment Is Impossible but Identification Still Succeeds **55**

33 Do-Calculus: The Complete Rules **56**

34 Non-Identifiability: When No Formula Exists **57**

35 Worked Examples **58**

Session 10: Causal Effects and Machine Learning **61**

36 From Identification to Estimation **61**
36.1 Outcome Regression 61
36.2 Propensity Scores and Inverse Propensity Weighting 61
36.3 Doubly Robust Estimation 62

37 Heterogeneous Effects and Meta-Learners **62**

38 Worked Examples **63**

Session 1: Foundations: Conditional Independence

Opening Motivation

Suppose you want to build a full probabilistic model of $n = 50$ binary symptoms and diseases for a diagnostic system. The joint distribution $P(X_1, \ldots, X_{50})$ has $2^{50} - 1$ free parameters — more numbers than there are seconds since the Big Bang. No dataset, no matter how large, could ever estimate them, and no computer could store them. Yet doctors reason about hundreds of symptoms every day. The resolution is not that the doctor secretly knows 2^{50} numbers; it is that most symptoms are *irrelevant to each other once you know the diagnosis.* This session makes that intuition precise. Conditional independence is the single idea that turns intractable joint distributions into tractable, learnable, and interpretable models — and it is the concept every remaining session in this course builds on.

1 Why the Joint Distribution Alone Is Not Enough

For discrete random variables $X_1, \ldots, X_n$, the chain rule of probability lets us write *any* joint distribution, with no assumptions at all, as

$$P(x_1, \ldots, x_n) = \prod_{i=1}^{n} P\big(x_i \mid x_1, \ldots, x_{i-1}\big).$$

This is always true, and it is almost always useless: the i-th factor is a table indexed by all $i - 1$ preceding variables, so the total number of free parameters is still $O(2^n)$ for binary variables. The chain rule tells us a joint distribution *can* be factored; it says nothing about *when the factors are small.* Figure 1 makes the blow-up concrete: with only ten binary variables, a full joint table already needs over a thousand parameters, while a chain-structured model with the same variables needs fewer than twenty.

The gap between these two curves is entirely explained by conditional independence (CI) assumptions. This is the organizing idea of the whole course: *a graphical model is nothing more than a compact way of writing down which conditional independencies hold, and a computationally efficient way of using them.* Everything from Session 2 onward is machinery built on top of this one idea.

2 Independence and Conditional Independence

Definition 2.1 (Independence)**.** Random variables X and Y are *independent*, written $X \perp\!\!\!\perp Y$, if

$$P(x, y) = P(x)P(y) \quad \text{for all values } x, y,$$

equivalently $P(x \mid y) = P(x)$ whenever $P(y) > 0$: learning Y tells you nothing about X.

Independence is rare in practice — almost everything is at least weakly related to everything else. The far more useful and far more common structure is independence *given some context.*

Definition 2.2 (Conditional Independence)**.** Let X, Y, Z be (possibly vector-valued) random variables. X and Y are *conditionally independent given* Z, written $X \perp\!\!\!\perp Y \mid Z$, if

$$P(x, y \mid z) = P(x \mid z)\, P(y \mid z) \quad \text{for all } x, y, z \text{ with } P(z) > 0.$$

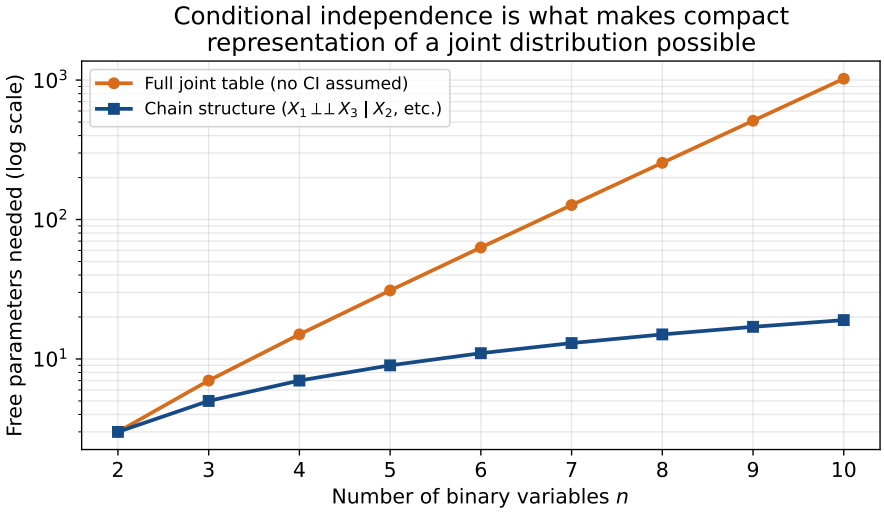


Figure 1: Free parameters required to specify a joint distribution over n binary variables: full joint table versus a chain structure exploiting conditional independence. Note the log scale.

Intuition

Think of Z as "using up" all of the statistical relationship between X and Y. Figure 2 shows this with overlapping regions representing shared information. On the left, X and Y overlap directly: knowing X tells you something about Y that nothing else explains. On the right, X and Y still overlap, but the *entire* overlap sits inside Z — once you already know Z, there is nothing left of the X–Y relationship to learn.

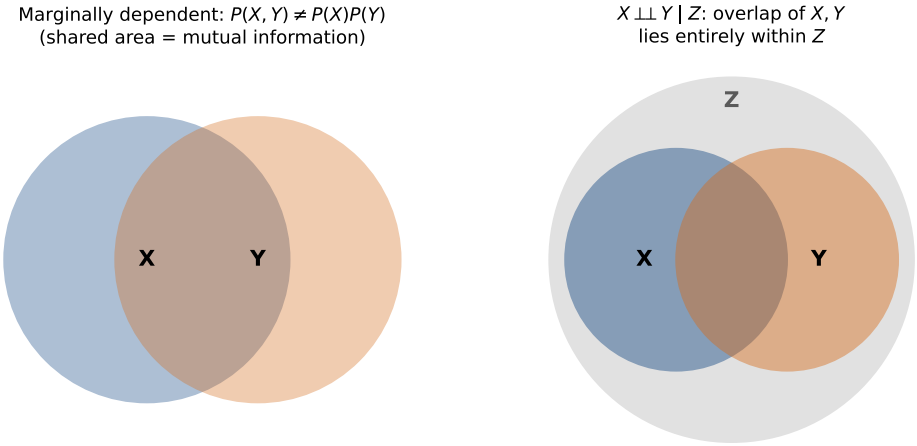


Figure 2: Left: X and Y share information not explained by anything else. Right: $X \perp\!\!\!\perp Y \mid Z$ — all shared information between X and Y routes through Z.

An equivalent and often more convenient characterization avoids explicitly normalizing:

Proposition 2.3 (Factorized form of CI)**.** *$X \perp\!\!\!\perp Y \mid Z$ if and only if there exist functions g and h*

such that

$$P(x, y, z) = g(x, z)\, h(y, z) \quad \textit{for all } x, y, z.$$

Proof sketch. ($\Rightarrow$) Take $g(x, z) = P(x \mid z)P(z)$ and $h(y, z) = P(y \mid z)$; then $g(x, z)h(y, z) = P(x \mid z)P(y \mid z)P(z) = P(x, y \mid z)P(z) = P(x, y, z)$ using Definition 2.2. ($\Leftarrow$) Sum both sides over x to get $P(y, z) = h(y, z)\sum_x g(x, z)$, and similarly for $P(x, z)$; substitute back to recover Definition 2.2. Full algebra in the appendix. □

This factorized view is exactly why CI matters computationally: it says the three-way table $P(x, y, z)$ can be built from two smaller tables g and h instead of one large one — a preview of the factor-based computation in Session 3.

2.1 Three Structural Properties

CI is not an arbitrary relation; it obeys algebraic rules that will let us manipulate independence statements symbolically, the way we manipulate equations.

Proposition 2.4 (Properties of conditional independence)**.** *For disjoint sets of variables X, Y, Z, W:*

(i) ***Symmetry:*** $X \perp\!\!\!\perp Y \mid Z \iff Y \perp\!\!\!\perp X \mid Z$.

(ii) ***Decomposition:*** $X \perp\!\!\!\perp (Y, W) \mid Z \implies X \perp\!\!\!\perp Y \mid Z$.

(iii) ***Weak union:*** $X \perp\!\!\!\perp (Y, W) \mid Z \implies X \perp\!\!\!\perp Y \mid (Z, W)$.

These are straightforward consequences of Definition 2.2 (see appendix for derivations) but they matter practically: decomposition says a CI statement about a group of variables implies CI for any subgroup, while weak union says that once we know X is irrelevant to the pair (Y, W), it stays irrelevant to Y even after we additionally condition on W. In Session 2, an entire graphical calculus (d-separation) will exist precisely to let us read off consequences like these directly from a picture, without redoing this algebra every time.

3 Worked Examples

Example 3.1 (Symptoms given a diagnosis)**.** A clinic models three symptoms — fever S_1, cough S_2, fatigue S_3 — and a binary diagnosis $D \in \{\text{flu}, \text{no flu}\}$. Suppose the modeler assumes each symptom is generated independently once the diagnosis is fixed:

$$S_1 \perp\!\!\!\perp S_2 \perp\!\!\!\perp S_3 \mid D$$

(shorthand for every pair being CI given D, and jointly $P(s_1, s_2, s_3 \mid d) = \prod_{k=1}^{3} P(s_k \mid d)$; see Figure 3). This is the classical *naive Bayes* assumption. Without it, modeling $P(s_1, s_2, s_3 \mid d)$ requires $2^3 - 1 = 7$ parameters per value of d; with it, only 3 parameters per value of d are needed — one per symptom. The assumption is almost certainly not perfectly true (a cough may make fatigue more likely even after conditioning on flu status), but it is often close enough to be highly useful, and it is learnable from a small dataset where the exact joint would not be.

Example 3.2 (Verifying CI numerically)**.** Let $Z \in \{0, 1\}$ be a fair coin, and let $X = Z \oplus N_1$, $Y = Z \oplus N_2$ where N_1, N_2 are independent fair coin flips (all independent of Z), and $\oplus$ is addition mod 2. Marginally, X and Y are dependent: $P(X{=}1) = P(Y{=}1) = 1/2$, but

$$P(X{=}1, Y{=}1) = P(Z{=}0)P(N_1{=}1)P(N_2{=}1) + P(Z{=}1)P(N_1{=}0)P(N_2{=}0) = \tfrac{1}{2} \cdot \tfrac{1}{4} + \tfrac{1}{2} \cdot \tfrac{1}{4} = \tfrac{1}{4},$$

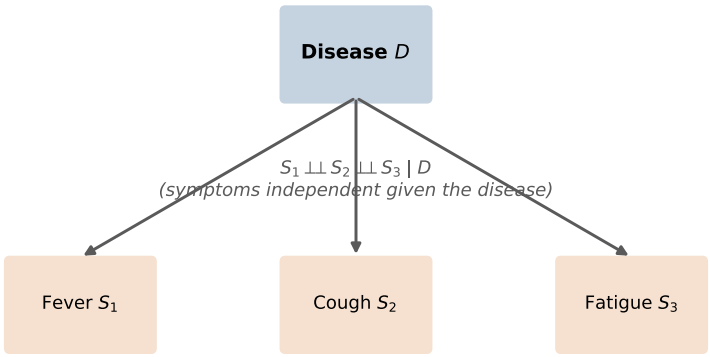


Figure 3: A shared cause D explains all correlation among the symptoms; conditioning on D leaves the symptoms independent.

which equals $P(X{=}1)P(Y{=}1) = 1/4$ here (this particular symmetric example happens to make them marginally independent too — a useful reminder that marginal independence and conditional independence are logically unrelated in general). Now condition on $Z = 0$: $X = N_1$ and $Y = N_2$, which are independent by construction, so $P(x, y \mid Z{=}0) = P(x \mid Z{=}0)P(y \mid Z{=}0)$ for every x, y; the same holds for $Z = 1$. Hence $X \perp\!\!\!\perp Y \mid Z$ holds exactly, by Definition 2.2, even though one cannot see this by staring at X and Y's marginal behavior alone — it is a statement that only becomes checkable once you fix Z.

Worked Problem 3.3 (Marginal vs. conditional independence can point opposite ways)**.** Consider a recommendation system with $Z =$ "user is a movie enthusiast" (binary), $X =$ "user rated Movie A highly," $Y =$ "user rated Movie B highly." Suppose that, given enthusiast status, ratings for the two unrelated movies are independent: $X \perp\!\!\!\perp Y \mid Z$, with $P(X{=}1 \mid Z{=}1) = 0.9$, $P(X{=}1 \mid Z{=}0) = 0.1$, and identically for Y; enthusiasts make up 30% of users, so $P(Z{=}1) = 0.3$. Show that X and Y are *marginally dependent* despite being conditionally independent.

Solution. By the law of total probability,

$$P(X{=}1, Y{=}1) = \sum_z P(X{=}1 \mid z)P(Y{=}1 \mid z)P(z) = (0.9)^2(0.3) + (0.1)^2(0.7) = 0.243 + 0.007 = 0.25.$$

Marginally $P(X{=}1) = 0.9(0.3) + 0.1(0.7) = 0.34$, so if X, Y were independent we would need $P(X{=}1, Y{=}1) = 0.34^2 = 0.1156$, far from the true value 0.25. The unobserved enthusiasm variable Z *induces* an apparent correlation between two otherwise unrelated ratings — a small preview of the confounding phenomena that will become central in Session 7.

Additional Worked Examples (Instructor Supplement)

Worked Problem 3.4 (Pairwise independence does not imply mutual independence — and flips under conditioning)**.** Let X, Y be independent fair coin flips, and let $Z = X \oplus Y$ (addition mod 2). We show all three pairs are *marginally* independent, yet $X \not\perp\!\!\!\perp Y \mid Z$ — a second, sharper illustration (beyond Definition 2.2's enthusiast example) of how marginal and conditional independence can point in opposite directions.

Marginal independence of every pair. By symmetry $P(X{=}1) = P(Y{=}1) = P(Z{=}1) = \frac{1}{2}$. Now $P(X{=}1, Z{=}1) = P(X{=}1, Y{=}0) = \frac{1}{4} = P(X{=}1)P(Z{=}1)$, so $X \perp\!\!\!\perp Z$; by an identical computation $Y \perp\!\!\!\perp Z$; and $X \perp\!\!\!\perp Y$ was assumed. So X, Y, Z are *pairwise* independent.

But not mutually independent. If X, Y, Z were mutually independent, we would need $P(X{=}1, Y{=}1, Z{=}1) = P(X{=}1)P(Y{=}1)P(Z{=}1) = \frac{1}{8}$. But $Z = X \oplus Y$, so $X{=}1, Y{=}1$ forces $Z{=}0$ with certainty: $P(X{=}1, Y{=}1, Z{=}1) = 0 \neq \frac{1}{8}$.

Conditioning on Z destroys the pairwise independence of X, Y. Given $Z = 0$, we know $Y = X$ exactly (since $X \oplus Y = 0 \iff X = Y$), so $P(X{=}1 \mid Y{=}1, Z{=}0) = 1 \neq P(X{=}1 \mid Z{=}0) = \frac{1}{2}$: knowing Y (given Z) tells you X with certainty. Hence $X \not\perp\!\!\!\perp Y \mid Z$, even though $X \perp\!\!\!\perp Y$ unconditionally. This is the mirror image of Session 1's Exercise 1.5 (there, conditional independence held but marginal independence failed via a confounder; here, marginal independence holds for every pair but conditioning on the third variable makes two of them perfectly dependent) — together the two examples show that no relationship of implication holds in either direction between $X \perp\!\!\!\perp Y$ and $X \perp\!\!\!\perp Y \mid Z$ in general.

Worked Problem 3.5 (A fully worked naive Bayes classification)**.** Return to the naive Bayes assumption from the disease-symptom example above (symptoms conditionally independent given diagnosis), now with a concrete email spam filter: class $C \in \{\text{spam}, \text{ham}\}$, and two word-presence indicators W_1, W_2 with $W_1 \perp\!\!\!\perp W_2 \mid C$. Suppose $P(C{=}\text{spam}) = 0.4$, $P(W_1{=}1 \mid \text{spam}) = 0.7$, $P(W_1{=}1 \mid \text{ham}) = 0.1$, $P(W_2{=}1 \mid \text{spam}) = 0.6$, $P(W_2{=}1 \mid \text{ham}) = 0.2$. An incoming email has both words present. By the CI assumption, $P(w_1, w_2 \mid c) = P(w_1 \mid c)P(w_2 \mid c)$, so

$$P(W_1{=}1, W_2{=}1 \mid \text{spam}) = 0.7 \times 0.6 = 0.42, \qquad P(W_1{=}1, W_2{=}1 \mid \text{ham}) = 0.1 \times 0.2 = 0.02.$$

By Bayes' rule, the unnormalized posteriors are $0.4 \times 0.42 = 0.168$ (spam) and $0.6 \times 0.02 = 0.012$ (ham), which sum to 0.18, giving

$$P(\text{spam} \mid w_1, w_2) = \frac{0.168}{0.18} \approx \mathbf{0.933}, \qquad P(\text{ham} \mid w_1, w_2) = \frac{0.012}{0.18} \approx 0.067.$$

Without the CI assumption, classifying on two words would require the full joint table $P(w_1, w_2 \mid c)$ — 3 free numbers per class instead of 2 — a small saving here, but one that becomes decisive with thousands of vocabulary words: the naive Bayes assumption is what makes text classification with a huge vocabulary statistically and computationally feasible at all.

Worked Problem 3.6 (Verifying weak union with explicit numbers)**.** Proposition 2.3's appendix proof of weak union is entirely algebraic; here is a concrete numerical check. Let $Z \in \{0, 1\}$ with $P(Z{=}0) = P(Z{=}1) = 0.5$, and suppose $X \perp\!\!\!\perp (Y, W) \mid Z$ (the weak-union hypothesis), with X's conditional distribution given only by $P(X{=}1 \mid Z{=}0) = 0.3$, and (Y, W)'s *joint* conditional distribution given $Z{=}0$ specified directly (and, notably, *correlated*: $P(Y{=}1) = 0.5 \neq 0.4 =$ their product, given below) by

$$P(Y{=}1, W{=}1 \mid Z{=}0) = 0.4, \;\; P(Y{=}1, W{=}0 \mid Z{=}0) = 0.1, \;\; P(Y{=}0, W{=}1 \mid Z{=}0) = 0.2, \;\; P(Y{=}0, W{=}0 \mid Z{=}0) = 0$$

Weak union claims $X \perp\!\!\!\perp Y \mid (Z, W)$ should now hold. Check it at $Z{=}0, W{=}1$: $P(W{=}1 \mid Z{=}0) = 0.4 + 0.2 = 0.6$, so $P(Y{=}1 \mid Z{=}0, W{=}1) = 0.4/0.6 = 2/3$. Since $X \perp\!\!\!\perp (Y, W) \mid Z$ by hypothesis, X does not depend on W given Z either (summing the hypothesis over Y), so $P(X{=}1 \mid Z{=}0, W{=}1) = P(X{=}1 \mid Z{=}0) = 0.3$. Now compute the joint: $P(X{=}1, Y{=}1, W{=}1 \mid Z{=}0) = P(X{=}1 \mid Z{=}0)\, P(Y{=}1, W{=}1 \mid Z{=}0) = 0.3 \times 0.4 = 0.12$, so $P(X{=}1, Y{=}1 \mid Z{=}0, W{=}1) = 0.12/0.6 = 0.2$. Compare to the product $P(X{=}1 \mid Z{=}0, W{=}1)\, P(Y{=}1 \mid Z{=}0, W{=}1) = 0.3 \times \frac{2}{3} = 0.2$. They match exactly, confirming $X \perp\!\!\!\perp Y \mid (Z, W)$ numerically, even though Y and W are themselves correlated given Z — weak union asks nothing about the relationship between Y and W, only that X stays irrelevant to Y once Z and W are both fixed.

Appendix: Derivations Omitted Above

Proof of Proposition 2.3, full algebra. For the forward direction, from Definition 2.2, $P(x,y,z) = P(z)P(x,y \mid z) = P(z)P(x \mid z)P(y \mid z)$. Group as $\big[P(x \mid z)P(z)\big] \cdot \big[P(y \mid z)\big] = g(x,z)h(y,z)$. For the converse, if $P(x,y,z) = g(x,z)h(y,z)$, marginalizing over x gives $P(y,z) = h(y,z)G(z)$ where $G(z) = \sum_x g(x,z)$, and marginalizing over y gives $P(x,z) = g(x,z)H(z)$ where $H(z) = \sum_y h(y,z)$. Marginalizing over both gives $P(z) = G(z)H(z)$. Then

$$P(x,y \mid z) = \frac{g(x,z)h(y,z)}{G(z)H(z)} = \frac{g(x,z)}{G(z)} \cdot \frac{h(y,z)}{H(z)} = P(x \mid z)P(y \mid z),$$

using $P(x \mid z) = P(x,z)/P(z) = g(x,z)/G(z)$ and similarly for $P(y \mid z)$.

Proof of symmetry, decomposition, weak union. Symmetry is immediate from Definition 2.2 since multiplication commutes. For decomposition, sum both sides of $P(x,y,w \mid z) = P(x \mid z)P(y,w \mid z)$ over w to get $P(x,y \mid z) = P(x \mid z)P(y \mid z)$. For weak union, from $P(x,y,w \mid z) = P(x \mid z)P(y,w \mid z)$, divide both sides by $P(w \mid z)$ (assuming positivity) to get $P(x,y \mid z,w) = P(x \mid z)\frac{P(y,w|z)}{P(w|z)}$; since the right side does not depend on y except through the second factor, and summing over y shows that factor equals $P(x \mid z,w)/P(x \mid z)$ is impossible unless it simplifies to $P(x \mid z,w)$ directly, one obtains $P(x \mid z,w) = P(x \mid z)$, from which $P(x,y \mid z,w) = P(x \mid z,w)P(y \mid z,w)$ follows after substitution. (Students may find it faster to verify weak union directly from the defining equation of decomposable graphs in Session 2, where it becomes a one-line graphical fact.)

Connections and Takeaways

This session established that (1) the chain rule always factors a joint distribution, but only conditional independence makes the factors small; (2) $X \perp\!\!\!\perp Y \mid Z$ has an equivalent product-factorization form (Proposition 2.3) that foreshadows how we will compute with factors rather than full tables; and (3) CI obeys algebraic rules (symmetry, decomposition, weak union) that let us reason about many independence statements at once.

What we do *not* yet have is a way to specify or read off *which* conditional independencies hold among n variables without writing out a wall of $\perp\!\!\!\perp$ statements by hand. **Session 2** answers this with a single idea: draw a directed graph whose missing edges encode conditional independence assumptions, and prove that a simple graphical test (d-separation) tells you exactly which CI statements the graph implies — turning today's algebra into a picture you can read at a glance.

Session 2: Bayesian Networks: DAGs and Independence Encoding

Opening Motivation

Last session we established that conditional independence (CI) is what makes a joint distribution compact, and we practiced deriving CI algebraically from properties like symmetry and decomposition. That algebra works, but it does not scale: for a model with fifty variables, writing out and manually verifying every relevant CI statement by hand is unrealistic. What we need is a *representation* — something we can draw — from which every CI statement the model assumes can be read off by inspection. Today's answer is the Bayesian network: a directed acyclic graph (DAG) whose missing edges silently encode a whole family of conditional independencies, and a graphical test, d-separation, that lets us read them off at a glance.

4 From Chain Rule to Graph

Recall the (always valid but generally useless) chain-rule factorization $P(x_1, \dots, x_n) = \prod_i P(x_i \mid x_1, \dots, x_{i-1})$. A Bayesian network is what you get when you are willing to *drop* some of the conditioning variables in each factor — keeping only a small, chosen subset $\mathrm{pa}(X_i)$ (the *parents* of X_i) for each variable, and asserting that this is enough.

Definition 4.1 (Bayesian network)**.** A Bayesian network is a pair (G, Θ) where G is a directed acyclic graph (DAG) with nodes $X_1, \dots, X_n$, and Θ specifies a conditional probability distribution $P(x_i \mid \mathrm{pa}(x_i))$ for every node. The network defines a joint distribution via the **chain rule for Bayesian networks**:

$$P(x_1, \dots, x_n) = \prod_{i=1}^{n} P\big(x_i \mid \mathrm{pa}(x_i)\big).$$

This factorization is a genuine assumption, called the **local Markov assumption**: each variable is conditionally independent of its non-descendants given its parents, $X_i \perp\!\!\!\perp \mathrm{nd}(X_i) \mid \mathrm{pa}(X_i)$. The DAG's edges are exactly the conditioning relationships the modeler chose to keep; every *missing* edge is an implicit claim of irrelevance.

Example 4.2 (The student-grades network)**.** Consider five variables familiar from any classroom: course **D**ifficulty (easy/hard), student **I**ntelligence (low/high), the **G**rade earned, the student's **S**AT score, and the recommendation **L**etter quality. A modeler encodes these causal-looking assumptions: Grade depends on Difficulty and Intelligence; SAT depends only on Intelligence; the Letter depends only on the Grade. Figure 4 draws this. The chain rule for this network gives

$$P(d, i, g, s, l) = P(d)\, P(i)\, P(g \mid d, i)\, P(s \mid i)\, P(l \mid g),$$

five small factors instead of one table over all $2 \times 2 \times 3 \times 2 \times 2 = 48$ joint configurations (assuming G has three grade levels) built from a single 47-parameter table.

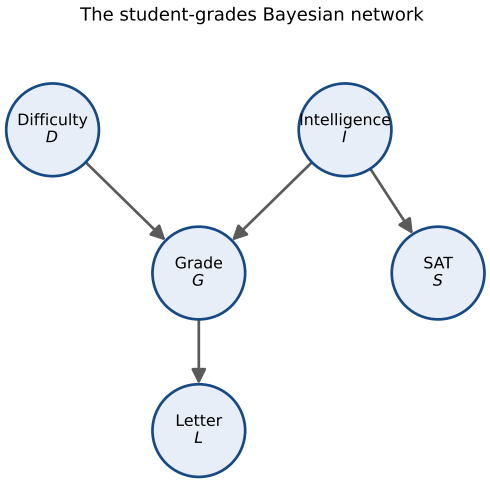


Figure 4: The student-grades Bayesian network. Missing edges (e.g. no edge from D to S) are the model's independence assumptions.

5 Reading Independence off the Graph: d-Separation

The local Markov assumption directly gives us *some* independencies (each node from its non-descendants given its parents), but it does not obviously tell us about, say, $D \perp\!\!\!\perp S$ (are course difficulty and SAT score independent, unconditionally?). We need a general graphical procedure that answers *any* query of the form "is $X \perp\!\!\!\perp Y \mid Z$ implied by this graph?" The key is to understand how probabilistic influence flows along paths, one edge-triple at a time.

5.1 Three Building Blocks

Every path in a DAG is built from three possible local patterns at each intermediate node, shown in Figure 5.

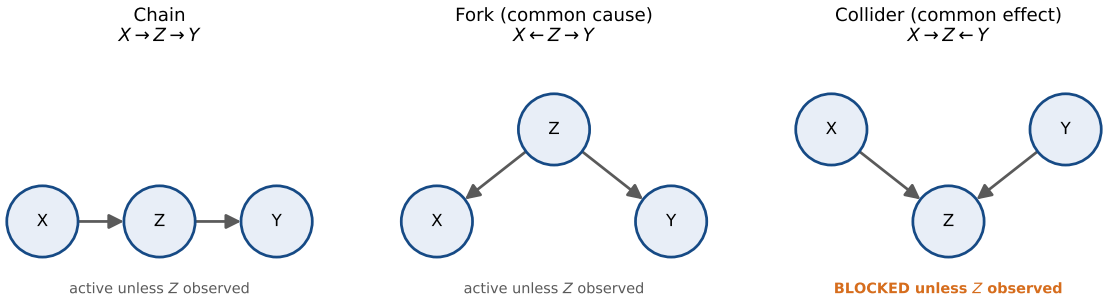


Figure 5: The three canonical connection patterns. Chains and forks transmit influence unless the middle node is observed; colliders do the opposite.

- **Chain** $X \to Z \to Y$: X influences Y only *through* Z. If we observe Z, that channel is cut: $X \perp\!\!\!\perp Y \mid Z$. If Z is unobserved, influence flows freely.

- **Fork** $X \leftarrow Z \rightarrow Y$: Z is a common cause; it creates a spurious correlation between X and Y. Again, conditioning on Z blocks the path: $X \perp\!\!\!\perp Y \mid Z$.
- **Collider** $X \rightarrow Z \leftarrow Y$: X and Y are marginally independent *causes* of a common effect Z — exactly like Session 1's worked example with Z the enthusiast indicator. Here the logic flips: with Z unobserved, the path is blocked and $X \perp\!\!\!\perp Y$; but observing Z (or any descendant of Z) *opens* the path, inducing dependence ("explaining away").

Definition 5.1 (d-separation)**.** A path between X and Y is *blocked* by a set of nodes Z if it contains a chain or fork node in Z, or a collider node such that neither the collider nor any of its descendants is in Z. Sets X and Y are *d-separated* by Z, written $X \perp\!\!\!\perp_d Y \mid Z$, if Z blocks *every* path between them.

Theorem 5.2 (Soundness of d-separation)**.** *If $X \perp\!\!\!\perp_d Y \mid Z$ in DAG G, then $X \perp\!\!\!\perp Y \mid Z$ in* every *distribution P that factorizes according to G.*

The proof (appendix) proceeds by induction on the three local patterns above, combined with the CI algebra from Session 1 (decomposition, weak union): d-separation is nothing more than a systematic bookkeeping of chain/fork/collider logic applied along every path at once. A converse also holds — if X and Y are *not* d-separated given Z, then $X \not\perp\!\!\!\perp Y \mid Z$ for *almost every* choice of numerical parameters Θ consistent with G (a property called *faithfulness*); we state this without proof, since only soundness (Theorem 5.2) is needed to trust that a graph never claims an independence that isn't really there.

Intuition

d-separation is best learned by tracing paths with your finger, not by memorizing the definition. Ask, for each path between X and Y: "does this path pass through a fork or chain node that IS shaded (observed)?" $\rightarrow$ blocked. "Does it pass through a collider that is NOT shaded, and none of whose descendants are shaded?" $\rightarrow$ blocked. Otherwise the path is active. $X \perp\!\!\!\perp_d Y \mid Z$ requires *every* path to be blocked — one active path is enough to destroy independence.

6 Worked Examples

Worked Problem 6.1 (Is $D \perp\!\!\!\perp S$?)**.** In the student network (Figure 4), is Difficulty d-separated from SAT with no evidence, $Z = \emptyset$?

Solution. The only path between D and S is $D \rightarrow G \leftarrow I \rightarrow S$. The intermediate node G is a collider on this path (two arrowheads meet at G), and $G \notin Z = \emptyset$, and no descendant of G (namely L) is in Z either. An unobserved collider blocks a path, so this single path is blocked, and since it is the only path, $D \perp\!\!\!\perp_d S \mid \emptyset$. Difficulty and SAT score are independent — consistent with the intuitive causal story that they share no cause and no evidence has been observed to make them "compete" as explanations of anything.

Worked Problem 6.2 (Conditioning on a collider creates dependence)**.** Now suppose we *do* observe the grade, $Z = \{G\}$. Is $D \perp\!\!\!\perp_d S \mid G$? See Figure 6.

Solution. The same path $D \rightarrow G \leftarrow I \rightarrow S$ now has its collider G observed ($G \in Z$). An observed collider *opens* the path rather than blocking it. Since this is the only path and it is now active, $D \not\perp\!\!\!\perp S \mid G$. Intuitively: once you know a student got a good grade, learning the course was

easy makes it *less* likely the student is highly intelligent (the "easy" explanation partially substitutes for the "smart" explanation) — which is exactly a statement about S (a proxy for intelligence) becoming informative about D. This "explaining away" phenomenon is a direct generalization of Session 1's enthusiast example, and it is also the graphical seed of an important warning: naively "controlling for everything you can measure" can *create* spurious associations rather than remove them — a theme we revisit sharply in Session 7.

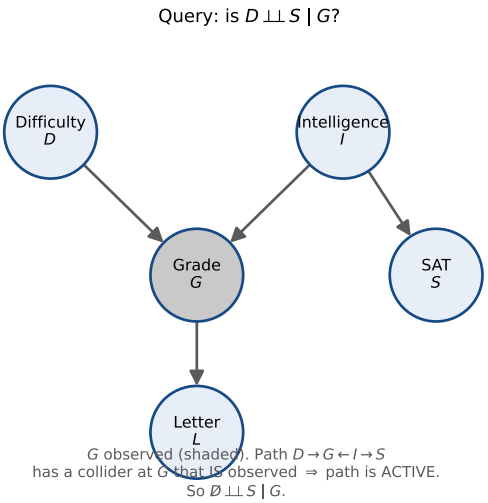


Figure 6: Observing the collider G (shaded) opens the previously blocked path between D and S.

Worked Problem 6.3 (Local Markov property in the medical network)**.** Recall Session 1's disease-symptom model, now drawn as a DAG with edges $D \to S_1$, $D \to S_2$, $D \to S_3$ (a fork at D with three children). Use d-separation to confirm $S_1 \perp\!\!\!\perp S_2 \mid D$.

Solution. The only path between S_1 and S_2 is $S_1 \leftarrow D \to S_2$, a fork with middle node D. Since $D \in Z = \{D\}$, the fork is blocked, so $S_1 \perp\!\!\!\perp_d S_2 \mid D$, confirming by the graphical criterion exactly the naive-Bayes assumption we asserted by hand (via algebra) in Session 1. This is the payoff promised at the top of this session: what took an explicit CI definition and direct verification last time is now a two-second visual check.

Additional Worked Examples (Instructor Supplement)

Worked Problem 6.4 (Conditioning on a *descendant* of a collider also opens the path)**.** The d-separation definition above says a collider blocks a path unless *the collider or one of its descendants* is in Z — the worked examples above only ever conditioned on the collider itself. Consider Rain R, Sprinkler S (independent causes), WetGrass G (a collider: $R \to G \leftarrow S$), and SlipperyFloor F, a further effect $G \to F$. Is $R \perp\!\!\!\perp_d S \mid F$?

Solution. The only path is $R \to G \leftarrow S$, with collider G. Here $Z = \{F\}$: $G \notin Z$, but F *is* a descendant of G, and $F \in Z$. By that definition, a collider blocks a path only if *neither* it nor any of its descendants is in Z — that condition fails here, so the path is **active**, and $R \not\perp\!\!\!\perp S \mid F$. Intuitively: learning the floor is slippery makes rain and sprinkler use *both* more plausible explanations, but if you additionally learn it was *not* raining, the sprinkler becomes more likely to have been on — exactly the explaining-away logic of Exercise 2.2, now triggered by evidence on a downstream

effect rather than on the collider itself. This is the clause students most often forget when applying d-separation, and it is exactly why post-treatment variables are dangerous to condition on — a warning Session 9's backdoor criterion will make precise.

Worked Problem 6.5 (A second network for practice: alarm, burglary, earthquake)**.** Consider the classic network: Burglary B and Earthquake E (independent, unconditionally) both cause Alarm A ($B \to A \leftarrow E$, a collider), and A causes two calls, JohnCalls J and MaryCalls M ($A \to J$, $A \to M$). Answer, with justification: (a) $B \perp\!\!\!\perp_d E \mid \emptyset$? (b) $B \perp\!\!\!\perp_d E \mid A$? (c) $J \perp\!\!\!\perp_d M \mid \emptyset$? (d) $J \perp\!\!\!\perp_d M \mid A$?

Solution. (a) The only path $B \to A \leftarrow E$ has collider $A \notin \emptyset$ and no descendant of A is in $\emptyset$ either, so the path is blocked: $B \perp\!\!\!\perp_d E \mid \emptyset$ — burglary and earthquakes are unconditionally unrelated, as expected. (b) Now $A \in Z = \{A\}$, opening the collider: $B \not\perp\!\!\!\perp E \mid A$ (if the alarm went off and it wasn't a burglary, an earthquake becomes more plausible — textbook explaining-away). (c) The only path between J and M is $J \leftarrow A \to M$, a *fork* with fork-node $A \notin \emptyset$, so it is unblocked: $J \not\perp\!\!\!\perp M \mid \emptyset$ (both calls are correlated simply because both depend on whether the alarm rang). (d) With $A \in Z$, the fork *is* now blocked: $J \perp\!\!\!\perp_d M \mid A$ — once you already know whether the alarm sounded, John's and Mary's calling behaviors carry no further information about each other. Notice (b) and (d) are opposite effects of the same act of conditioning on A: it opens the collider path for (B, E) while closing the fork path for (J, M) — the same node can play both roles simultaneously for different pairs of variables in a larger graph.

Worked Problem 6.6 (Deriving a chain's independence directly from the factorization)**.** For the simple chain $A \to B \to C$, derive $A \perp\!\!\!\perp C \mid B$ directly from the BN chain rule (Session 2's Definition 2.1), without invoking d-separation as a black box — reproducing, on a minimal example, exactly the kind of step the appendix's inductive proof of Theorem 5.2 performs in general.

Solution. The chain rule gives $P(a, b, c) = P(a)P(b \mid a)P(c \mid b)$ (since $\mathrm{pa}(A) = \emptyset$, $\mathrm{pa}(B) = \{A\}$, $\mathrm{pa}(C) = \{B\}$). Then

$$P(a, c \mid b) = \frac{P(a, b, c)}{P(b)} = \frac{P(a)P(b \mid a)P(c \mid b)}{P(b)} = \left(\frac{P(a)P(b \mid a)}{P(b)}\right) P(c \mid b) = P(a \mid b)\, P(c \mid b),$$

using Bayes' rule to identify the parenthesized term as $P(a \mid b)$. This is exactly Definition 1.2 of conditional independence, confirming $A \perp\!\!\!\perp C \mid B$ purely algebraically — and it is precisely Session 1's Proposition 1.2 (factorized form, $g(a, b)h(c, b)$ with $g(a, b) = P(a)P(b \mid a)$, $h(c, b) = P(c \mid b)$) applied to this specific factorization. D-separation's chain rule ("a chain node, once observed, blocks the path") is nothing more than this five-line computation, generalized and automated so it never needs to be redone by hand for every new graph.

Appendix: Proof Sketch of Theorem 5.2

The proof proceeds by strong induction on the topological structure of G. **Base case (local Markov property):** by construction of the BN factorization, each X_i is CI of its non-descendants given its parents; this is a direct algebraic consequence of the chain-rule factorization (sum out descendants and use the factorized form, Proposition 1.2 from Session 1, applied recursively from the "leaves" of the DAG upward). **Inductive step:** any CI statement $X \perp\!\!\!\perp_d Y \mid Z$ implied by blocking every path can be derived from a finite sequence of applications of the local Markov property together with the symmetry, decomposition, and weak-union properties established in Session 1 — each blocked chain or fork corresponds to

one weak-union/decomposition step, and each unblocked collider corresponds to a case where those properties do *not* apply, which is exactly where the induction stops extending independence. A complete formal treatment (Verma & Pearl, 1988) proceeds via the equivalent and sometimes more convenient notion of undirected separation in the *moralized ancestral graph* — take the smallest ancestral subgraph containing X, Y, Z, connect ("marry") any two parents sharing a child, drop edge directions, and test ordinary graph separation. We will reuse exactly this moralization idea from the opposite direction in Session 4, when relating Bayesian networks to Markov networks.

Connections and Takeaways

We now have a representation (the DAG) that compactly encodes a joint distribution's factorization, and an algorithm (d-separation) that reads off every conditional independence the graph implies — without ever touching the numbers in Θ. This closes the loop opened in Session 1: independence assumptions are no longer something we verify laboriously by algebra, but something we can see.

But notice what we have *not* yet done: we have never actually computed a probability like $P(I = \text{high} \mid L = \text{weak})$ from this network. Knowing the graph's independence structure is necessary for efficient computation, but it is not the computation itself. **Session 3** turns this structural knowledge into an algorithm — variable elimination — that computes exact answers to probability queries by systematically exploiting exactly the structure d-separation reveals, without ever building the full joint table we showed was intractable in Session 1.

Session 3: Inference: Variable Elimination

Opening Motivation

A doctor enters a patient's observed symptoms into a diagnostic Bayesian network and wants $P(\text{Disease} \mid \text{Symptoms observed})$. A streaming service has built a network relating latent taste clusters to observed ratings and wants $P(\text{likes genre} \mid \text{ratings so far})$. In both cases we have a Bayesian network — a graph plus local conditional probability tables, exactly what Session 2 gave us — but we have not yet asked the practical question: *how do we actually compute an answer?* The naive approach, summing over the full joint distribution, is exactly the $O(2^n)$ catastrophe Session 1 opened with. Today's algorithm, variable elimination, shows how to answer such queries *exactly*, while doing work that scales with the graph's structure rather than blindly with 2^n.

7 The Inference Problem

Definition 7.1 (Probabilistic query)**.** Given a Bayesian network over variables $\mathbf{X}$, a query variable set $Y \subset \mathbf{X}$, and observed evidence $E = e$ for some disjoint subset $E \subset \mathbf{X}$, *inference* means computing

$$P(y \mid e) = \frac{P(y, e)}{P(e)} = \frac{\sum_w P(y, e, w)}{\sum_{y', w} P(y', e, w)}, \qquad W = \mathbf{X} \setminus (Y \cup E),$$

where W are the remaining ("nuisance") variables summed out.

Computing $P(y, e)$ by summing the full joint table over all values of W is the naive algorithm; for $|W|$ binary variables it costs $O(2^{|W|})$. The Bayesian network factorization $P(\mathbf{x}) = \prod_i P(x_i \mid \text{pa}(x_i))$ from Session 2 gives us a smarter option: because the joint is a *product of small local factors*, we can push summations inside the product — summing out each nuisance variable as early as possible, working only with the factors that actually mention it.

8 Factors: The Currency of Computation

Definition 8.1 (Factor)**.** A *factor* $\phi(\mathbf{Y})$ over a scope $\mathbf{Y} \subseteq \mathbf{X}$ is any function assigning a non-negative real number to each joint assignment of $\mathbf{Y}$. Every conditional probability table $P(x_i \mid \text{pa}(x_i))$ is a factor with scope $\{X_i\} \cup \text{pa}(X_i)$; evidence is incorporated by simply restricting a factor to the observed value.

Two operations on factors are all variable elimination (VE) ever needs:

- **Factor product**: $(\phi_1 \cdot \phi_2)(\mathbf{Y}_1 \cup \mathbf{Y}_2) = \phi_1(\mathbf{y}_1)\phi_2(\mathbf{y}_2)$, matching shared variables.
- **Sum-out (marginalization)**: $\big(\sum_Z \phi\big)(\mathbf{Y} \setminus Z) = \sum_z \phi(\mathbf{y} \setminus Z, z)$.

8.1 The Variable Elimination Algorithm

Algorithm 1 Variable Elimination for computing $P(Y, e)$

1: Initialize the factor set Φ with every CPT of the network, each restricted to evidence $E = e$.
2: Choose an elimination order π over the nuisance variables $W = \mathbf{X} \setminus (Y \cup E)$.
3: **for** each variable Z in order π **do**
4: Collect all factors in Φ whose scope contains Z; call their product ψ.
5: Compute $\tau = \sum_Z \psi$ (sum Z out of ψ).
6: Remove the collected factors from Φ; add τ to Φ.
7: **end for**
8: Multiply all remaining factors in Φ to get an (unnormalized) factor over Y; normalize to sum to 1.

Figure 7 shows one elimination step in isolation: every factor that mentions the variable being eliminated is multiplied together first, and only then is the summation performed — this order of operations (multiply the relevant pieces, *then* sum) is the entire trick that avoids ever materializing the full joint table.

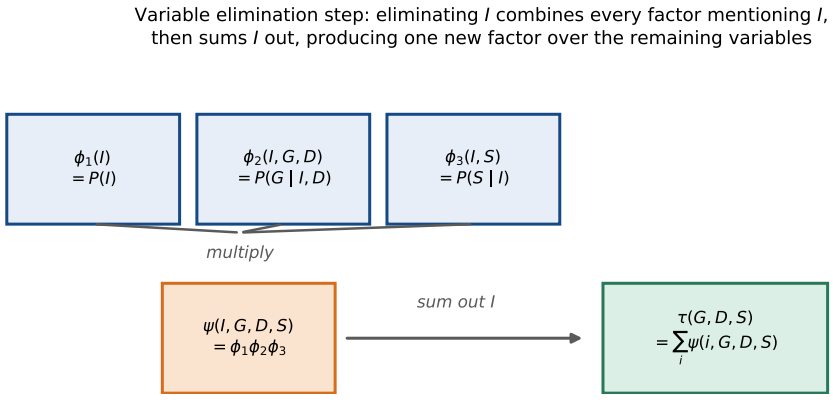


Figure 7: One step of variable elimination: combine every factor mentioning the eliminated variable, then marginalize.

Proposition 8.2 (Correctness of variable elimination)**.** *The value returned by variable elimination equals $P(y, e)$ exactly, for any elimination order.*

Proof idea. This follows from two facts: (1) summation distributes over multiplication, so pushing a sum for Z inside the product of only the factors that mention Z, and leaving the rest outside, changes nothing algebraically; (2) repeating this for every variable in W, one at a time, is exactly the process of computing $\sum_w \prod_i P(x_i \mid \mathrm{pa}(x_i))$ — the definition of $P(y, e)$ — just grouped efficiently. A full inductive proof is in the appendix. □

9 Elimination Order Matters: Treewidth

Correctness holds for *any* order, but cost does not. Each elimination step's cost is exponential in the number of variables in the largest factor *created*, since that factor's table has one entry per joint

assignment of its scope. Choosing a bad order can create huge intermediate factors even when a good order would not.

Definition 9.1 (Induced graph and treewidth)**.** Given an elimination order π, the *induced graph* adds an edge between any two variables that ever appear together in the same factor during elimination (this includes "fill-in" edges created when eliminating a variable connects all of its remaining neighbors). The *induced width* of π is the size of the largest clique in the induced graph, minus one. The *treewidth* of the graph is the minimum induced width over all elimination orders.

Proposition 9.2 (Complexity of VE)**.** *Variable elimination with order π of induced width w costs $O(n \cdot d^{w+1})$ time, where d is the largest variable domain size and n the number of variables.*

Figure 8 shows how sharply cost grows in w: this is the price of a bad ordering, and it is entirely avoidable structure — unlike the raw 2^n we started with in Session 1, which had no escape at all.

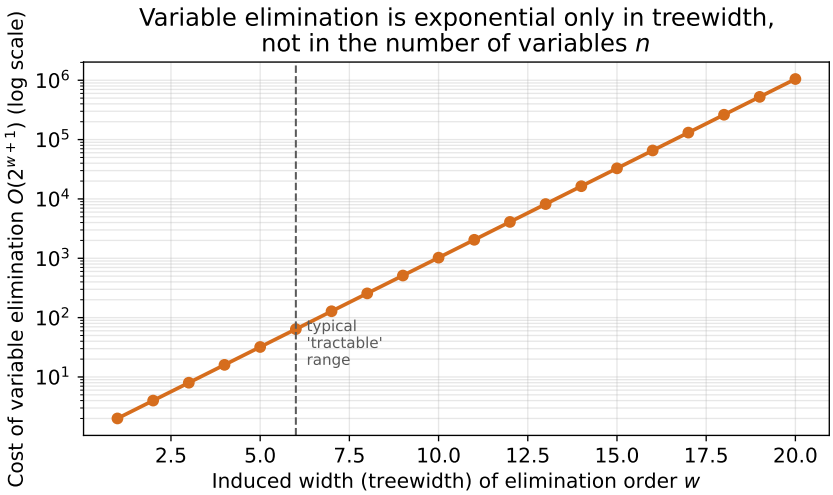


Figure 8: Variable elimination's cost is exponential in treewidth w, not in the number of variables n. Sparse, tree-like graphs (small w) stay tractable at any n.

Intuition

A good heuristic ("min-fill"): at each step, eliminate whichever remaining variable would add the fewest fill-in edges. Intuitively, eliminate leaves and near-leaves of the graph first, and save densely-connected "hub" variables for last — eliminating a hub early forces you to connect all of its many neighbors to each other, ballooning the factor size.

10 Worked Examples

Worked Problem 10.1 (Elimination order changes cost, not the answer)**.** Consider the moralized skeleton of the student network (Session 2): edges $D{-}I$ (married parents of G), $D{-}G$, $I{-}G$, $I{-}S$, $G{-}L$. Compare eliminating in order (L, S, G, D, I) versus eliminating I first.

Solution. See Figure 9. Eliminating L then S then G then D then I: L's only neighbor is G (cost $O(d^2)$, no fill-in); S's only neighbor is I (no fill-in); G's remaining neighbors are D, I, already connected (no fill-in, cost $O(d^3)$); the largest factor created has scope size 3, so induced width

$w = 2$. Eliminating I *first* instead: I's neighbors are D, G, S; none of D–S or G–S are edges yet, so eliminating I forces two new fill-in edges, creating a factor over $\{D, G, S\}$ of scope 3 immediately, and cascading complexity for the rest. Same final answer $P(y, e)$ either way (correctness doesn't depend on order) but a strictly worse induced width from the bad order.

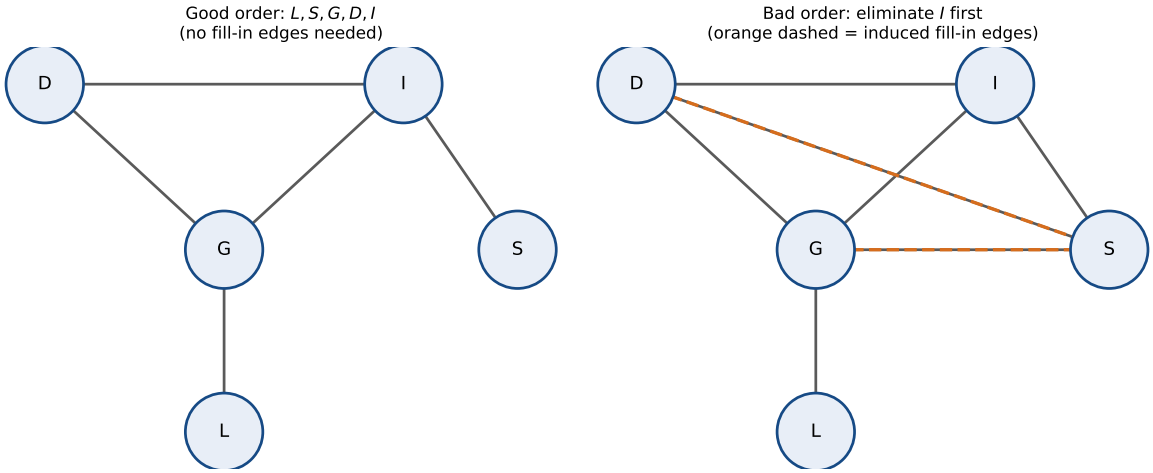


Figure 9: Left: eliminating leaves first needs no fill-in edges. Right: eliminating the well-connected node I first forces new (orange, dashed) edges among its neighbors.

Worked Problem 10.2 (Computing a query by hand)**.** Using the student network's factors $P(D)$, $P(I)$, $P(G \mid D, I)$, $P(S \mid I)$, $P(L \mid G)$, compute $P(I \mid L = \text{weak})$ symbolically via VE with order (S, D, G) eliminated before normalizing over I.

Solution. Step 1 (eliminate S): the only factor mentioning S is $P(S \mid I)$; summing over all values of S gives $\sum_s P(s \mid i) = 1$ for every i (a CPT's rows sum to one), so this factor simply vanishes — reflecting that SAT score is d-separated from the query given no other evidence touches it (exactly the collider logic of Session 2, since S is a leaf here). Step 2 (eliminate D): multiply $P(D)$ and $P(G \mid D, I)$, then sum over D: $\tau_1(G, I) = \sum_d P(d)P(G \mid d, I)$. Step 3 (eliminate G): multiply $\tau_1(G, I)$ by the evidence-restricted factor $P(L{=}\text{weak} \mid G)$, then sum over G: $\tau_2(I) = \sum_g \tau_1(g, I)\, P(L{=}\text{weak} \mid g)$. Finally, $P(I, L{=}\text{weak}) = P(I) \cdot \tau_2(I)$, and normalizing over the two values of I gives $P(I \mid L{=}\text{weak})$. Notice that S dropped out for free — a query never needs to touch variables that are d-separated from it given the evidence, which is exactly why Session 2's graphical reasoning and this session's computational reasoning are two views of the same fact.

Worked Problem 10.3 (A recommendation-system factor graph)**.** A streaming platform models three variables per session: a latent taste cluster $C \in \{1, \ldots, K\}$, a genre exposure indicator X (did the system show genre g), and an observed click Y. Factors: $\phi_1(C) = P(C)$, $\phi_2(C, X) = P(X \mid C)$, $\phi_3(C, X, Y) = P(Y \mid C, X)$ (click probability depends on both taste and what was shown). To compute $P(C \mid Y = 1)$, eliminate X first: $\tau(C, Y) = \sum_x \phi_2(C, x)\phi_3(C, x, Y)$; then normalize $\phi_1(C) \cdot \tau(C, Y{=}1)$ over C. This factor-graph view — factors need not correspond to a single "child given parents" CPT — generalizes the Bayesian network picture and is exactly the representation Session 4 will formalize for undirected models.

Additional Worked Examples (Instructor Supplement)

Worked Problem 10.4 (Factor product and sum-out, fully by hand)**.** Every worked example above described variable elimination structurally; here is one full numeric pass. Let $A \to B \to C$ with $\phi_1(A,B) = P(B \mid A)$ given by $\phi_1(0,0){=}0.6, \phi_1(0,1){=}0.4, \phi_1(1,0){=}0.3, \phi_1(1,1){=}0.7$, and $\phi_2(B,C) = P(C \mid B)$ given by $\phi_2(0,0){=}0.9, \phi_2(0,1){=}0.1, \phi_2(1,0){=}0.2, \phi_2(1,1){=}0.8$. Eliminate B: form $\psi(A,B,C) = \phi_1(A,B)\phi_2(B,C)$, then $\tau(A,C) = \sum_B \psi(A,B,C)$.

$$\tau(0,0) = \phi_1(0,0)\phi_2(0,0) + \phi_1(0,1)\phi_2(1,0) = 0.6(0.9) + 0.4(0.2) = 0.54 + 0.08 = \mathbf{0.62},$$

$$\tau(0,1) = 0.6(0.1){+}0.4(0.8) = 0.06{+}0.32 = \mathbf{0.38}, \qquad \tau(1,0) = 0.3(0.9){+}0.7(0.2) = 0.27{+}0.14 = \mathbf{0.41},$$

$$\tau(1,1) = 0.3(0.1) + 0.7(0.8) = 0.03 + 0.56 = \mathbf{0.59}.$$

As a sanity check, each row sums to 1 (0.62+0.38=1, 0.41+0.59=1), confirming $\tau(A,C) = P(C \mid A)$ exactly, since eliminating the middle node of a chain by factor product and sum-out is nothing but matrix multiplication of the two conditional probability tables — the general VE algorithm doing, mechanically, what we would otherwise compute by the law of total probability.

Worked Problem 10.5 (Chaining the computation to a full query)**.** Extend Exercise 10.4 with a fourth variable D, $C \to D$, $P(D{=}1 \mid C{=}0) = 0.3$, $P(D{=}1 \mid C{=}1) = 0.7$, and prior $P(A{=}1) = 0.5$. Compute $P(D{=}1)$ by continuing variable elimination: eliminate A next, then C. Eliminating A from $\tau(A,C)$: $P(C{=}0) = \sum_a P(a)\tau(a,0) = 0.5(0.62) + 0.5(0.41) = 0.31 + 0.205 = \mathbf{0.515}$, and $P(C{=}1) = 0.5(0.38) + 0.5(0.59) = 0.19 + 0.295 = \mathbf{0.485}$ (summing to 1, as required). Finally eliminate C: $P(D{=}1) = P(C{=}0)(0.3){+}P(C{=}1)(0.7) = 0.515(0.3){+}0.485(0.7) = 0.1545{+}0.3395 = \mathbf{0.494}$. This four-variable chain required only four small multiplications and additions at each step — never a table larger than two variables — exactly Proposition 3.2's cost guarantee, here realized concretely rather than asymptotically.

Worked Problem 10.6 (Trees always have treewidth 1)**.** Prove that *any* tree-structured graph (moralized or not, since a tree has no colliders with unmarried parents) has treewidth exactly 1, regardless of how many variables n it has.

Solution. Eliminate variables in a *leaf-first* order: at every step, some node of the current (shrinking) tree has at most one remaining neighbor (every tree with ≥ 2 nodes has at least one leaf). Eliminating a node with a single neighbor creates *no* fill-in edges at all (fill-in only connects *pairs* of a node's remaining neighbors, and a single neighbor has no partner to connect to), and after removal the remaining graph is still a tree (removing a leaf from a tree yields a tree). Repeating this $n-1$ times, every elimination step operates on a factor of scope at most 2 (the eliminated variable and its one neighbor), giving induced width $w = 1$ throughout. Since this holds for *every* n, variable elimination on a tree-structured model costs only $O(n \cdot d^2)$ — linear in the number of variables — no matter how large the tree is. This is the precise structural reason Figure 8's cost curve stays flat for sparse, tree-like graphs at any n, and it is the same leaf-first principle Exercise 3.1 used informally on the (non-tree, but still low-treewidth) student network.

Appendix: Full Inductive Proof of VE Correctness

We show by induction on $|W|$ that after processing the first k variables of W in order π, the remaining factor set Φ multiplies to $\sum_{w_1,\ldots,w_k} \prod_i P(x_i \mid \mathrm{pa}(x_i))|_{E=e}$ (summed over only the first k eliminated variables, with the rest of W and Y still free). **Base case** $k = 0$:

Φ is exactly the CPTs restricted to evidence, whose product is $P(\mathbf{x})|_{E=e}$, matching the claim trivially. **Inductive step**: assume the claim for k; at step $k+1$ we collect every factor in Φ whose scope contains Z_{k+1} into a product ψ, and the remaining factors Φ' do not depend on Z_{k+1}. By the inductive hypothesis, $\Phi \cup \Phi' = \psi \cdot (\text{product of } \Phi')$ equals $\sum_{w_1,\ldots,w_k} \prod_i P(x_i \mid \text{pa}(x_i))|_e$. Since Φ' factors do not depend on Z_{k+1}, they can be pulled outside a sum over Z_{k+1}: $\sum_{z_{k+1}} \psi \cdot \Phi' = \Phi' \cdot \sum_{z_{k+1}} \psi = \Phi' \cdot \tau$, exactly the new factor set after step $k+1$, matching the claim for $k+1$. After all of W is eliminated ($k = |W|$), Φ multiplies to $\sum_w \prod_i P(x_i \mid \text{pa}(x_i))|_e = P(y, e)$ by definition, completing the proof. Normalizing by $\sum_y P(y, e) = P(e)$ then yields $P(y \mid e)$.

Connections and Takeaways

Variable elimination gives us *exact* inference whose cost is governed entirely by the graph's treewidth — a purely structural quantity, computable before ever touching the numbers in Θ. This is a genuinely different kind of tool than Sessions 1–2: those sessions were about *representing* independence; this session is about *exploiting* it computationally.

But VE was presented entirely for directed Bayesian networks, using factors that happened to be conditional probability tables $P(x_i \mid \text{pa}(x_i))$. The factor-graph view in the last worked example hints at something more general: factors don't need a "child given parents" interpretation at all — they can just be arbitrary non-negative compatibility scores between variables that mutually constrain each other, with no natural direction of generation. **Session 4** formalizes exactly this: undirected graphical models (Markov networks), needed whenever the dependencies in a system are symmetric rather than generative — and variable elimination, as we'll see, applies to them with almost no modification at all.

Session 4: Markov Networks: Undirected Models

Opening Motivation

Consider modeling a photograph as a grid of pixels, where neighboring pixels tend to have similar brightness. Or a social network, where friends' opinions tend to align. In both cases there is a clear dependency structure, but no natural direction: pixel (i, j) does not "cause" pixel $(i, j+1)$'s brightness any more than the reverse, and my opinion influencing my friend's is just as true as the reverse. Forcing a DAG onto such symmetric relationships means arbitrarily picking an ordering and inventing a fictitious generative story — and, worse, as we will see today, some independence patterns literally cannot be represented by *any* DAG. Markov networks (also called Markov random fields) drop the directionality requirement entirely and represent dependency with plain undirected edges.

11 Gibbs Distributions

Definition 11.1 (Gibbs distribution / Markov network)**.** Given an undirected graph H over variables $X_1, \ldots, X_n$, a set of *potential functions* (or factors) $\phi_1, \ldots, \phi_m$, one per clique C_k (a fully-connected subset) of H, defines a joint distribution

$$P(x_1, \ldots, x_n) = \frac{1}{Z} \prod_{k=1}^{m} \phi_k(x_{C_k}), \qquad Z = \sum_{x_1, \ldots, x_n} \prod_{k=1}^{m} \phi_k(x_{C_k}),$$

where Z, the *partition function*, is a normalizing constant.

Unlike a Bayesian network's CPTs, a potential ϕ_k need not sum to one and need not represent "a probability of something given something else" — it is simply a non-negative compatibility score: how much the joint configuration x_{C_k} is favored, in relative terms. This is precisely the generalization the recommendation-system factor $\phi_3(C, X, Y)$ hinted at in Session 3's last worked example: factors as raw compatibility scores, not necessarily conditional probabilities.

Intuition

Think of $\log \phi_k$ as an "energy" contribution: configurations with high potential (low energy) are more probable, configurations with low potential are penalized, and Z simply rescales everything to sum to one. In an image-denoising grid (Figure 10), a pairwise potential $\phi_{ij}(x_i, x_j)$ that is large when $x_i = x_j$ and small otherwise encodes "neighboring pixels prefer to agree" — with no claim about which pixel came first.

The crucial computational fact: *variable elimination from Session 3 applies to Gibbs distributions with no modification at all.* VE only ever needed factors that could be multiplied and summed — it never used the fact that BN factors happened to be conditional probability tables. This is the payoff promised at the end of Session 3.

12 Reading Independence from an Undirected Graph

Markov networks have their own, and in fact *simpler*, independence-reading rule than d-separation — no colliders to worry about.

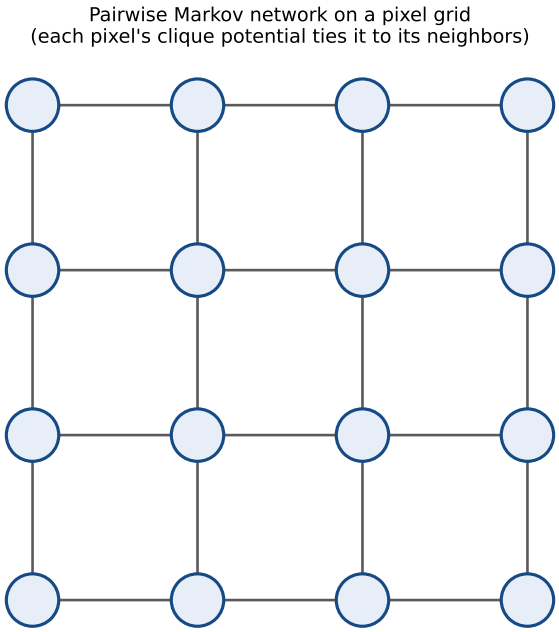


Figure 10: A pairwise Markov network on a pixel grid. Every edge carries a potential favoring agreement between neighbors; there is no natural direction to any edge.

Definition 12.1 (Global Markov property)**.** X and Y are separated by Z in undirected graph H, written $X \perp_H Y \mid Z$, if every path between X and Y passes through some node in Z (ordinary graph separation — remove Z and check if X, Y land in different connected components).

Theorem 12.2 (Hammersley–Clifford)**.** *If P is a strictly positive distribution ($P(\mathbf{x}) > 0$ for all $\mathbf{x}$), then P satisfies the global Markov property with respect to H if and only if P factorizes as a Gibbs distribution over the cliques of H.*

We state this landmark result without proof (it is a genuinely non-trivial argument via Möbius inversion; see the appendix for a sketch and references) because what matters for practice is simpler and more useful: *once* you have a Gibbs factorization over H's cliques, ordinary graph separation immediately tells you every conditional independence the model implies — no chain/fork/collider bookkeeping required. Positivity is the one condition worth remembering: without it (e.g., a distribution with hard logical constraints setting some configurations to probability exactly zero), the equivalence can fail.

13 Bayesian Networks and Markov Networks Are Incomparable

Every DAG has an undirected "shadow" that captures its independencies, but building it requires one extra step beyond simply dropping arrowheads.

Definition 13.1 (Moralization)**.** The *moral graph* of a DAG G is the undirected graph obtained by (1) connecting every pair of parents that share a common child ("marrying" them), and (2) dropping all edge directions.

Proposition 13.2. *Every distribution representable by a Bayesian network with DAG G is representable by a Markov network on the moral graph of G (set each clique potential equal to the*

corresponding CPT). The converse is not generally true.

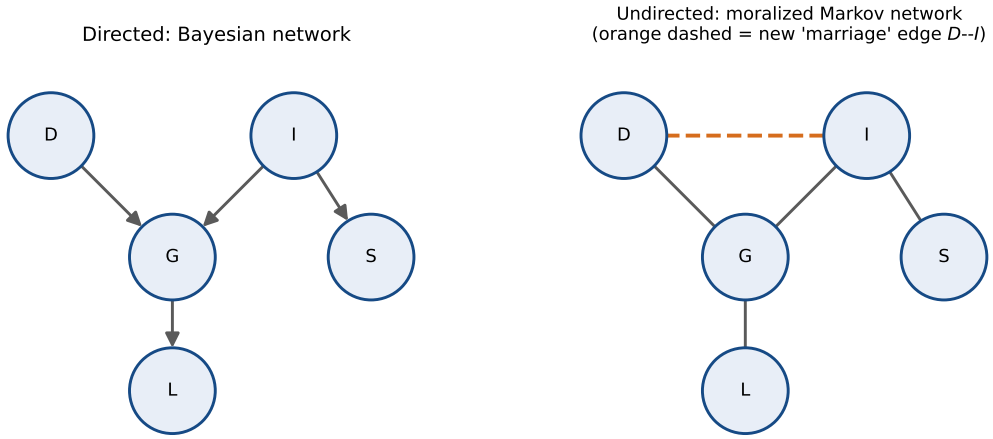


Figure 11: Moralizing the student network: parents D, I of the collider G are married with a new edge before dropping directions. Without this edge, ordinary separation would wrongly claim $D \perp I$ unconditionally, when actually D and I become dependent once G is observed (Session 2's explaining-away).

The marriage step is not bookkeeping trivia — it is forced by exactly the collider asymmetry from Session 2. An unmarried D–I pair sharing child G would let ordinary undirected separation claim $D \perp I \mid \emptyset$, true, but it would also (wrongly, if unmarried) claim $D \perp I \mid G$, when Session 2 showed explaining-away makes this false. Marrying parents of a common child is precisely what closes this gap.

But moralization is a one-way translation, and the families are genuinely incomparable:

- A collider $X \to Z \leftarrow Y$ encodes $X \perp\!\!\!\perp Y$ (marginally) but $X \not\perp\!\!\!\perp Y \mid Z$. No undirected graph on three nodes can express *both* facts simultaneously with the separation rule — any edge configuration is either "always separated by removing Z" or "never separated," but never asymmetric like this.
- Conversely, a 4-cycle A–B–C–D–A with no chord encodes $A \perp\!\!\!\perp C \mid \{B, D\}$ and $B \perp\!\!\!\perp D \mid \{A, C\}$ simultaneously. No DAG on four nodes reproduces exactly this independence pattern (any acyclic orientation of the cycle creates at least one collider, breaking the required symmetry).

Figure 12 summarizes this: the two model classes overlap (e.g., trees and chains can be written either way) but neither contains the other.

14 Worked Examples

Worked Problem 14.1 (Moralizing and reading independence)**.** Moralize the student network and verify $S \perp\!\!\!\perp L \mid I, G$ using ordinary separation.

Solution. The moral graph adds edge D–I (marrying G's parents) and keeps D–G, I–G, I–S, G–L (Figure 11). Removing $\{I, G\}$ from the graph: S's only edge was to I (now removed), so S becomes isolated; L's only edge was to G (also removed), so L becomes isolated too. S and L

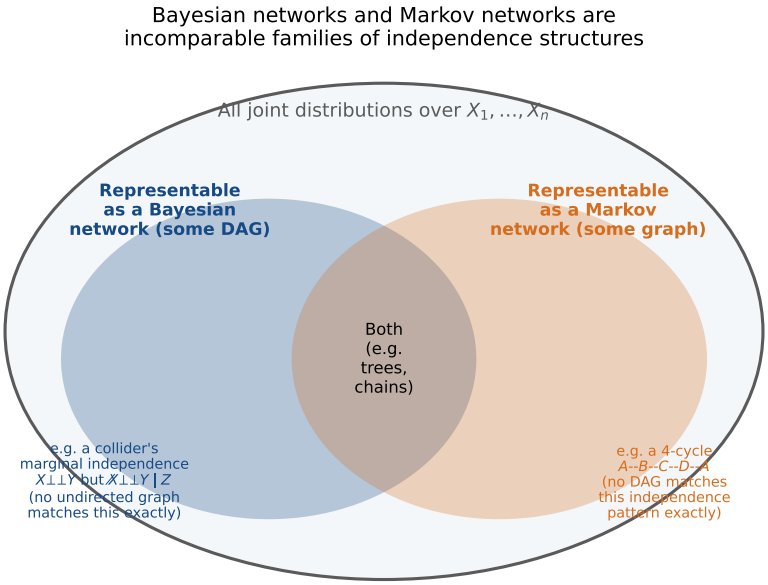


Figure 12: Bayesian networks and Markov networks are complementary, not nested, families of representable independence structures.

land in different (trivial, single-node) components, so $S \perp_H L \mid \{I, G\}$ — matching the DAG's own d-separation answer, as it must, since moralization is designed to preserve every independence the DAG implies (the converse direction — Markov networks implying strictly *more* independencies than moralization shows — is exactly the incomparability point above).

Worked Problem 14.2 (A 4-cycle with no DAG equivalent)**.** Four collaborating scientists A, B, C, D arranged in a review cycle (A reviews B's work, B reviews C's, C reviews D's, D reviews A's) each subtly bias their own opinions toward whichever colleague reviewed them, but only directly adjacent scientists influence each other. Model this as a Markov network with edges A–B, B–C, C–D, D–A and pairwise potentials $\phi(x_i, x_j)$ favoring agreement. By separation, $A \perp C \mid \{B, D\}$ and $B \perp D \mid \{A, C\}$. Attempt to draw a DAG with the same four nodes reproducing exactly these two facts: any acyclic assignment of directions to the 4-cycle's edges creates at least one node with two incoming edges from non-adjacent-in-the-cycle nodes (a collider), which by Session 2 would make that pair dependent given the collider rather than independent — contradicting one of the two required statements. This confirms the model families are genuinely different, not just different notations for the same thing.

Worked Problem 14.3 (Computing with Gibbs potentials via VE)**.** For the 2×2 image-denoising grid with pixels $X_{11}, X_{12}, X_{21}, X_{22} \in \{0, 1\}$ and pairwise potentials $\phi(x_i, x_j) = e^{x_i x_j}$ (encouraging agreement) on each of the four grid edges, compute Z by variable elimination: eliminate X_{22} first (appears in two potentials, with X_{12} and X_{21}), multiply those two potentials and sum over $X_{22} \in \{0, 1\}$ to get a new factor $\tau(X_{12}, X_{21})$; continue eliminating X_{21}, then X_{12}, then X_{11}, multiplying whatever factors remain at each step. The mechanics are identical, term for term, to Session 3's VE on a Bayesian network — only the interpretation of the factors (compatibility scores rather than CPTs) and the presence of an explicit Z to normalize at the very end have changed.

Additional Worked Examples (Instructor Supplement)

Worked Problem 14.4 (Computing an actual Gibbs distribution, start to finish)**.** Let $X_1, X_2 \in \{0,1\}$ with a single pairwise potential $\phi(x_1, x_2) = e^{x_1 x_2}$ (only the joint configuration $(1,1)$ is rewarded). Compute the full normalized distribution.

Solution. $\phi(0,0) = \phi(0,1) = \phi(1,0) = e^0 = 1$, and $\phi(1,1) = e^1 \approx 2.71828$. The partition function is $Z = 1 + 1 + 1 + 2.71828 = 5.71828$. Normalizing,

$$P(1,1) = \frac{2.71828}{5.71828} \approx \mathbf{0.4754}, \qquad P(0,0) = P(0,1) = P(1,0) = \frac{1}{5.71828} \approx \mathbf{0.1749} \text{ each},$$

which indeed sum to 1 ($0.4754 + 3(0.1749) = 0.4754 + 0.5247 \approx 1$). This is the full mechanics Definition 4.1 describes only symbolically: a raw compatibility score for each configuration, entirely un-normalized on its own, becomes a genuine probability distribution only after dividing by Z — and note $\phi(1,1)$ need not (and does not) sum to 1 over anything by itself, unlike a CPT.

Worked Problem 14.5 (Numerically confirming why moralization must marry the parents of a collider)**.** Consider $D \to G \leftarrow I$ with $P(D{=}1) = P(I{=}1) = 0.5$ (independent) and $P(G{=}1 \mid D, I)$ given by $P(G{=}1 \mid 0,0){=}0.1$, $P(G{=}1 \mid 0,1){=}0.9$, $P(G{=}1 \mid 1,0){=}0.9$, $P(G{=}1 \mid 1,1){=}0.95$. If moralization *skipped* marrying D, I, undirected separation would (incorrectly) claim $D \perp I \mid G$. Check this numerically.

$$P(D{=}1, I{=}1, G{=}1) = 0.25(0.95) = 0.2375, \quad P(D{=}1, I{=}0, G{=}1) = 0.25(0.9) = 0.225,$$

$$P(D{=}0, I{=}1, G{=}1) = 0.25(0.9) = 0.225, \quad P(D{=}0, I{=}0, G{=}1) = 0.25(0.1) = 0.025,$$

so $P(G{=}1) = 0.2375 + 0.225 + 0.225 + 0.025 = 0.7125$. Then $P(D{=}1 \mid G{=}1) = (0.2375 + 0.225)/0.7125 = 0.4625/0.7125 \approx 0.6491$, and by the symmetry of this particular CPT, $P(I{=}1 \mid G{=}1) \approx 0.6491$ too. If $D \perp I \mid G$ held, we would need $P(D{=}1, I{=}1 \mid G{=}1) = 0.6491 \times 0.6491 \approx 0.4213$. But the true value is $P(D{=}1, I{=}1 \mid G{=}1) = 0.2375/0.7125 \approx \mathbf{0.3333}$ — nowhere near 0.4213. The gap (0.3333 vs. 0.4213) is exactly the numerical fingerprint of explaining-away, and it is precisely why the moral graph *must* include the married edge D–I: without it, ordinary graph separation would confidently predict a wrong number, not just a qualitatively wrong independence claim.

Appendix: Hammersley–Clifford Proof Sketch

The forward direction (Gibbs ⇒ global Markov) follows the same style of argument as Session 2's d-separation soundness proof: factor the Gibbs distribution into the piece depending on one side of a separator Z and the piece depending on the other, exactly mirroring Session 1's Proposition 1.2 (the $g(x,z)h(y,z)$ factorized-CI form) generalized to cliques. The converse (global Markov ⇒ Gibbs, requiring positivity) is proved via the Möbius inversion formula: define $\phi_C(\mathbf{x}) = \prod_{S \subseteq C} \big[-\log P(\mathbf{x}_S, \mathbf{x}_{-S} = 0) \big]^{(-1)^{|C|-|S|}}$ for each clique C, relative to an arbitrary reference configuration $\mathbf{0}$, and shows algebraically that these reconstruct P exactly, using positivity to guarantee $\log P$ is always well-defined. See Koller & Friedman, *Probabilistic Graphical Models* (2009), Theorem 4.1, for the complete argument; it is a beautiful but genuinely technical result that we intentionally do not reprove line by line in-session.

Connections and Takeaways

We now have two complementary graphical languages — directed (Session 2) and undirected (today) — each encoding independence via a different local pattern-matching rule, both computable exactly with the same variable-elimination machinery (Session 3). Between them, essentially any independence structure a domain expert can describe is representable by *some* graphical model.

Every session so far, however, made an unstated assumption: that we already *know* the graph G (or H) and the numerical parameters Θ that go with it. Where do these come from in practice? **Session 5** tackles the numbers first: given a fixed graph structure and a dataset, how do we estimate Θ by maximum likelihood — and what happens when some of the variables in our carefully-drawn graph are never actually observed in the data at all?

Session 5: Learning: Parameter Estimation and Latent Variables

Opening Motivation

Every model in Sessions 2–4 assumed the numbers were simply handed to us: $P(G \mid D, I)$, clique potentials, and so on. In reality, these numbers come from data. If we are lucky, every variable in our graph is observed in every data record, and estimating parameters is almost mechanical — a direct payoff of the factorization we spent two sessions establishing. But often the most useful variables in a model, like a customer's underlying "segment" or a patient's true (versus tested) disease status, are never directly observed. Today we handle both cases: maximum likelihood estimation when everything is observed, and the EM algorithm — built directly on Session 3's inference machinery — when it is not.

15 Maximum Likelihood Estimation with Full Observability

Suppose we are given the DAG G of a Bayesian network and a dataset $\mathcal{D} = \{\mathbf{x}^{(1)}, \ldots, \mathbf{x}^{(N)}\}$ of N fully-observed i.i.d. samples. We want $\hat{\theta} = \arg\max_\theta \; P(\mathcal{D} \mid \theta)$.

Proposition 15.1 (Decomposable log-likelihood)**.** *For a Bayesian network with parameters $\theta = (\theta_1, \ldots, \theta_n)$, where θ_i parameterizes $P(x_i \mid \mathrm{pa}(x_i); \theta_i)$, the log-likelihood decomposes as*

$$\ell(\theta; \mathcal{D}) = \log \prod_{j=1}^{N} \prod_{i=1}^{n} P\big(x_i^{(j)} \mid \mathrm{pa}(x_i)^{(j)}; \theta_i\big) = \sum_{i=1}^{n} \underbrace{\sum_{j=1}^{N} \log P\big(x_i^{(j)} \mid \mathrm{pa}(x_i)^{(j)}; \theta_i\big)}_{\ell_i(\theta_i; \mathcal{D})} .$$

This is an immediate consequence of the DAG factorization from Session 2, combined with log turning products into sums. Its significance is entirely practical: if the parameter sets $\theta_1, \ldots, \theta_n$ do not overlap (the usual case), *maximizing $\ell(\theta; \mathcal{D})$ decomposes into n separate, independent optimization problems*, one per node, each involving only that node and its parents.

Consequence: closed-form CPT estimates. For discrete CPTs, each $\ell_i(\theta_i)$ is maximized (subject to each row of the CPT summing to one, via a Lagrange multiplier) at the empirical conditional frequency:

$$\hat{P}(x_i = v \mid \mathrm{pa}(x_i) = u) = \frac{\mathrm{count}(X_i{=}v, \mathrm{pa}(X_i){=}u)}{\mathrm{count}(\mathrm{pa}(X_i){=}u)}.$$

□

Intuition

Learning a Bayesian network with full observability reduces to counting: for every node, tally how often each value occurred alongside each parent configuration, and normalize. This is only possible *because* the graph told us, via the local Markov property, exactly which variables are relevant to each CPT — a smaller graph means fewer counts to accumulate and less data needed per parameter, directly mirroring Session 1's compactness argument.

Worked Problem 15.2 (Counting CPTs for the student network)**.** Suppose 200 fully-observed student records give: among the 80 students with difficulty=hard, 32 have intelligence=high; among the 120 with difficulty=easy, 60 have intelligence=high. (Suppose D and I are modeled as marginally independent, so we only need $P(I)$, not $P(I \mid D)$, matching Figure 2.2's graph.) Then $\hat{P}(I{=}\text{high}) = (32+60)/200 = 0.46$. For $P(G \mid D, I)$, we would similarly tally grade outcomes within each of the four (D, I) combinations and normalize each of those four rows separately — exactly Proposition 15.1's claim that each CPT is estimated independently of every other one.

16 When Observability Fails: Latent Variables

Now suppose some variables are never recorded. A marketing team observes purchase amounts but never directly asks customers which underlying "segment" (budget-conscious, aspirational, luxury) they belong to; a language model sees words but never sees the latent topic that generated them. Call the observed variables X and the latent (hidden) variables C; Figure 13 shows this structure using *plate notation*, a compact way to draw "repeat this substructure N times."

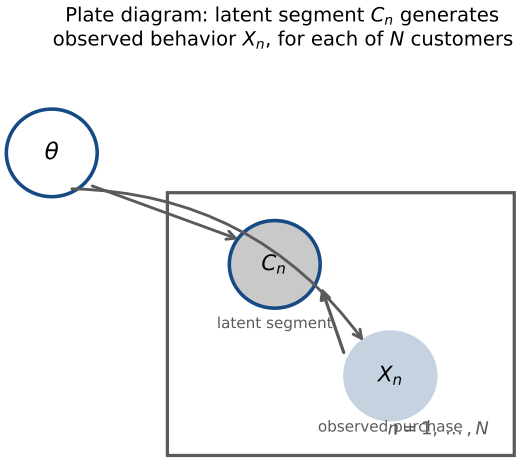


Figure 13: Plate notation: the box means "one copy of $C_n \to X_n$ for each of $n = 1, \ldots, N$ customers," with θ shared across all copies. Shaded/filled nodes are observed; the unfilled-gray node C_n is latent.

The problem: Proposition 15.1's decomposition assumed we could plug $\mathbf{x}^{(j)}$ directly into every factor. With C unobserved, the likelihood of the data we *can* see requires summing C out:

$$\ell(\theta; \mathcal{D}) = \sum_{j=1}^{N} \log P(x^{(j)}; \theta) = \sum_{j=1}^{N} \log \sum_{c} P(x^{(j)}, c; \theta),$$

and the log of a *sum* no longer decomposes into separate per-node terms the way Proposition 15.1 did. This is not a minor inconvenience — it usually makes direct maximization analytically intractable.

17 The EM Algorithm

The Expectation-Maximization (EM) algorithm sidesteps this by alternating between (1) using the *current* parameter estimate to infer a distribution over the missing C given what is observed — exactly a Session-3 inference query — and (2) treating those inferred (soft) values as if they were observed, reducing back to the easy, decomposable MLE problem from the previous section.

Algorithm 2 Expectation–Maximization

1: Initialize $\theta^{(0)}$ arbitrarily (e.g., randomly).
2: **for** $t = 0, 1, 2, \ldots$ until convergence **do**
3: **E-step:** for each data point j, compute $q_j(c) := P(c \mid x^{(j)}; \theta^{(t)})$ (an inference query — Session 3's VE).
4: **M-step:** set $\theta^{(t+1)} = \arg\max_\theta \sum_j \sum_c q_j(c) \log P(x^{(j)}, c; \theta)$ (an ordinary weighted MLE, decomposable exactly as in Proposition 15.1, now with "soft counts" $q_j(c)$ replacing observed indicator values).
5: **end for**

Theorem 17.1 (EM monotonicity)**.** *The sequence of observed-data log-likelihoods produced by EM is non-decreasing: $\ell(\theta^{(t+1)}; \mathcal{D}) \geq \ell(\theta^{(t)}; \mathcal{D})$ for every t.*

The proof (appendix) uses Jensen's inequality to show the M-step maximizes a lower bound on ℓ that is guaranteed to touch ℓ exactly at $\theta^{(t)}$ — so any improvement to the bound is a genuine (or at worst equal) improvement to the true log-likelihood. EM is not guaranteed to reach a global maximum (it can and does get stuck in local optima, so multiple random restarts are standard practice), but Theorem 17.1 guarantees it never gets *worse*, which is the essential property that makes it a reliable, widely-used algorithm rather than a heuristic.

18 Worked Examples

Worked Problem 18.1 (EM by hand: one iteration on a tiny mixture)**.** Two data points $x^{(1)} = 2$, $x^{(2)} = 7$ are believed to come from a two-component Gaussian mixture with known equal variance $\sigma^2 = 1$ and known mixing weights $\pi_0 = \pi_1 = 0.5$, but unknown means, initialized at $\mu_0^{(0)} = 3$, $\mu_1^{(0)} = 6$. *E-step for $x^{(1)} = 2$:* unnormalized responsibilities $\propto \exp(-\frac{1}{2}(2-3)^2) = e^{-0.5} \approx 0.607$ for component 0, and $\propto \exp(-\frac{1}{2}(2-6)^2) = e^{-8} \approx 0.00034$ for component 1; normalizing, $q_1(c{=}0 \mid x^{(1)}) \approx 0.9994$, $q_1(c{=}1 \mid x^{(1)}) \approx 0.0006$ — point 1 is confidently assigned to component 0. Symmetric computation for $x^{(2)} = 7$ gives $q_2(c{=}1 \mid x^{(2)}) \approx 0.9994$. *M-step:* the new mean for component 0 is a responsibility-weighted average, $\mu_0^{(1)} = \frac{q_1(0)\cdot 2 + q_2(0)\cdot 7}{q_1(0)+q_2(0)} \approx \frac{0.9994(2)+0.0006(7)}{1.0} \approx 2.003$, barely moved from the true value because the assignment was already almost certain; similarly $\mu_1^{(1)} \approx 6.997$. Even from a mediocre initialization, one iteration nearly separates the two components correctly here because the points were far apart relative to σ.

Worked Problem 18.2 (Customer segmentation via EM)**.** A firm has $N = 400$ customers' purchase amounts, believed to arise from two latent segments with different average spend, but the segment label was never recorded. Running EM (initializing means at two random data points) for 25 iterations on this data gives the log-likelihood trajectory in Figure 14 — monotonically increasing at every step exactly as Theorem 17.1 guarantees — converging to recovered segment parameters

close to the true (simulated) generating parameters, shown against the raw data histogram in Figure 15. The striking part: the algorithm was never told there were two segments' worth of *labels*, only that there might be two components, and it recovered both the mixing proportions and the two per-segment spending distributions purely from the shape of the unlabeled data.

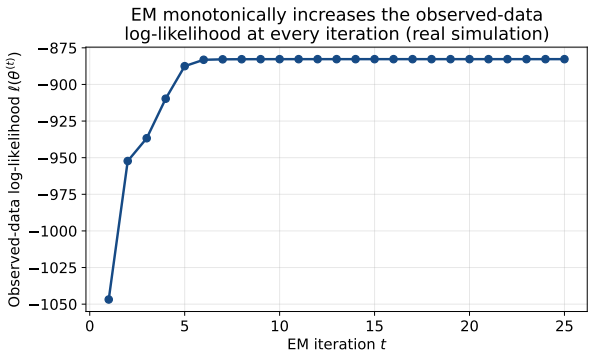


Figure 14: Real EM run: log-likelihood increases monotonically every iteration.

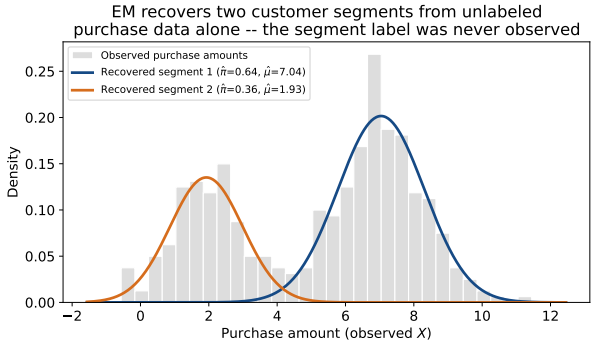


Figure 15: Recovered two-segment mixture overlaid on the observed (unlabeled) purchase data.

Worked Problem 18.3 (Why the E-step is "just" Session 3's inference)**.** Suppose the mixture above is instead written as a two-node Bayesian network $C \to X$ (segment causes purchase behavior), with $P(C)$ the mixing weights and $P(X \mid C)$ the per-segment spending distribution — exactly the plate diagram in Figure 13 unrolled for one customer. The E-step's $q_j(c) = P(c \mid x^{(j)}; \theta^{(t)})$ is then *literally* the query "$P(C \mid X{=}x^{(j)})$" from a two-node network with current parameters $\theta^{(t)}$ — solvable in one line by Bayes' rule (a trivial case of Session 3's variable elimination, since there is only one nuisance-free query variable and one piece of evidence). For larger latent-variable networks, the E-step is exactly VE with the current $\theta^{(t)}$ plugged in; nothing new needs to be learned computationally, only combined with the M-step's ordinary weighted MLE.

Additional Worked Examples (Instructor Supplement)

Worked Problem 18.4 (A second fully-observed counting example)**.** A two-node network $A \to B$ has $N = 10$ fully-observed records: $(A, B) = (1,1) \times 4,\ (1,0) \times 1,\ (0,1) \times 2,\ (0,0) \times 3$. Compute the full MLE CPTs.

Solution. $\hat{P}(A{=}1) = 5/10 = 0.5$ (five records have $A{=}1$: the four $(1,1)$'s and one $(1,0)$). Among those five, $B{=}1$ occurs 4 times, so $\hat{P}(B{=}1 \mid A{=}1) = 4/5 = 0.8$. Among the five records with $A{=}0$, $B{=}1$ occurs 2 times, so $\hat{P}(B{=}1 \mid A{=}0) = 2/5 = 0.4$. Exactly as Proposition 15.1 guarantees, each CPT row was estimated purely by counting within its own parent configuration, with no reference to the other rows at all.

Worked Problem 18.5 (Deriving the Bernoulli MLE by calculus)**.** The main text asserts the counting formula follows "via a Lagrange multiplier"; here is the calculation for the simplest case, a single Bernoulli(θ) node with i.i.d. data $x_1, \ldots, x_N \in \{0, 1\}$.

Solution. $\ell(\theta) = \sum_{i=1}^{N} \big[x_i \log\theta + (1 - x_i)\log(1-\theta)\big]$. Differentiating and setting to zero:

$$\frac{d\ell}{d\theta} = \sum_i \left[\frac{x_i}{\theta} - \frac{1 - x_i}{1 - \theta}\right] = 0 \implies (1-\theta)\sum_i x_i = \theta \sum_i (1 - x_i) = \theta N - \theta \sum_i x_i,$$

$$\implies \sum_i x_i - \theta \sum_i x_i = \theta N - \theta \sum_i x_i \implies \sum_i x_i = \theta N \implies \hat{\theta} = \frac{1}{N} \sum_i x_i,$$

the sample mean — exactly the empirical frequency used in every counting formula in this session, now derived from first-order calculus rather than merely asserted. (The second derivative is negative wherever $0 < \theta < 1$, confirming this critical point is indeed the maximum.)

Worked Problem 18.6 (EM with a genuinely ambiguous point)**.** Three points $x = (1, 5, 9)$ come from a two-component mixture, $\sigma^2 = 1$, mixing weights 0.5/0.5, initialized at $\mu_0^{(0)} = 2, \mu_1^{(0)} = 8$. Unlike Exercise 5.1 (where both points were confidently assigned), $x = 5$ sits exactly midway.

E-step. For x=1: unnormalized responsibilities $\propto e^{-\frac{1}{2}(1-2)^2} = e^{-0.5} \approx 0.607$ (component 0) vs. $e^{-\frac{1}{2}(1-8)^2} = e^{-24.5} \approx 0$ (component 1), so $q(0 \mid 1) \approx 1$. Symmetrically $q(1 \mid 9) \approx 1$. For x=5: $e^{-\frac{1}{2}(5-2)^2} = e^{-4.5}$ for *both* components (since $|5-2| = |5-8| = 3$), so the responsibilities are exactly **tied**: $q(0 \mid 5) = q(1 \mid 5) = 0.5$.

M-step. $\mu_0^{(1)} = \dfrac{q(0 \mid 1)(1) + q(0 \mid 5)(5) + q(0 \mid 9)(9)}{q(0 \mid 1) + q(0 \mid 5) + q(0 \mid 9)} = \dfrac{1(1) + 0.5(5) + 0(9)}{1 + 0.5 + 0} = \dfrac{3.5}{1.5} \approx \mathbf{2.333}$,

and by symmetry $\mu_1^{(1)} = \dfrac{0(1) + 0.5(5) + 1(9)}{0 + 0.5 + 1} = \dfrac{11.5}{1.5} \approx \mathbf{7.667}$. The ambiguous point $x = 5$ pulls *both* means slightly toward the center (from $2 \to 2.333$ and $8 \to 7.667$) by contributing a genuine half-share to each — this is exactly what "soft" assignment means in practice: EM does not have to commit an uncertain point to one cluster, unlike a hard-assignment algorithm (e.g., k-means), which would be forced to break the tie arbitrarily.

Appendix: Proof of EM Monotonicity (Theorem 17.1)

Define the expected complete-data log-likelihood under the current posterior, $Q(\theta \mid \theta^{(t)}) = \sum_j \sum_c q_j^{(t)}(c) \log P(x^{(j)}, c; \theta)$, where $q_j^{(t)}(c) = P(c \mid x^{(j)}; \theta^{(t)})$. One can show

$$\ell(\theta; \mathcal{D}) - Q(\theta \mid \theta^{(t)}) + \sum_j \sum_c q_j^{(t)}(c) \log q_j^{(t)}(c) = \sum_j \mathrm{KL}\big(q_j^{(t)}(\cdot) \,\|\, P(\cdot \mid x^{(j)}; \theta)\big) \geq 0,$$

by Jensen's inequality applied to the concavity of log (this rearrangement is the standard "evidence lower bound" or ELBO decomposition). At $\theta = \theta^{(t)}$, the KL term is exactly zero (since $q_j^{(t)}$ is defined as that very posterior), so $\ell(\theta^{(t)}; \mathcal{D}) = Q(\theta^{(t)} \mid \theta^{(t)}) - \sum_j \sum_c q_j^{(t)}(c) \log q_j^{(t)}(c)$. Since the M-step chooses $\theta^{(t+1)} = \arg\max_\theta Q(\theta \mid \theta^{(t)})$, we have $Q(\theta^{(t+1)} \mid \theta^{(t)}) \geq Q(\theta^{(t)} \mid \theta^{(t)})$, and since KL divergence is always non-negative, $\ell(\theta^{(t+1)}; \mathcal{D}) \geq Q(\theta^{(t+1)} \mid \theta^{(t)}) - \sum_j \sum_c q_j^{(t)}(c) \log q_j^{(t)}(c) \geq Q(\theta^{(t)} \mid \theta^{(t)}) - \sum_j \sum_c q_j^{(t)}(c) \log q_j^{(t)}(c) = \ell(\theta^{(t)}; \mathcal{D})$, completing the proof.

Connections and Takeaways

We can now learn CPT parameters exactly when data is fully observed (a direct, decomposable payoff of the DAG factorization), and approximately-optimally even with latent variables, by alternating inference (Session 3) with the same decomposable MLE. Two loose threads remain before this toolkit is genuinely practical. First, we have always assumed the graph structure G itself is known — but where does the graph come from when domain

knowledge is incomplete? Second, the E-step's inference call assumed variable elimination is tractable, but Session 3 showed VE's cost explodes with treewidth — what happens when it is not tractable? **Session 6** addresses both: a brief look at learning structure from data, and a full treatment of sampling-based approximate inference (including Markov chain Monte Carlo) for exactly the cases where exact inference is computationally out of reach.

Session 6: Structure Learning and Approximate Inference

Opening Motivation

Two loose threads remain from Session 5. First, every session so far assumed the graph itself — who is a parent of whom, which pairs of variables share an edge — was simply handed to us by a domain expert. What if it isn't? Second, the EM algorithm's E-step leaned entirely on Session 3's variable elimination, but VE's cost is exponential in treewidth — what happens in the many realistic models (dense social networks, image grids, large medical ontologies) where treewidth is too large for exact inference to run in any reasonable time? This session closes both threads: briefly, for structure; at length, via sampling, for inference. The sampling ideas developed here — especially Markov chain Monte Carlo (MCMC) — will also turn out, in Session 10, to be the computational backbone of modern causal effect estimation.

19 Learning Structure: Why It's Hard, and What's Done in Practice

Proposition 19.1 (The DAG space explodes)**.** *The number of distinct DAGs on n labeled nodes grows super-exponentially in n (Robinson's recurrence, appendix); already at $n = 8$ nodes there are over 7.8×10^{11} possible DAGs (Figure 16).*

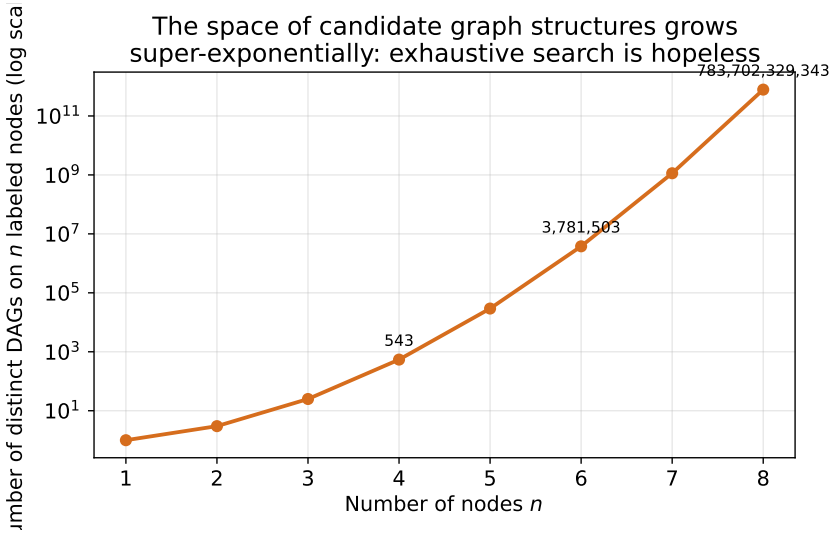


Figure 16: The number of candidate DAG structures explodes far faster than the number of variables. Exhaustively scoring every structure is infeasible beyond a handful of nodes.

Two broad families of practical structure-learning methods exist:

- **Score-based search**: define a score (e.g., BIC, which penalizes log-likelihood by $\frac{\log N}{2}$ times the number of parameters, directly generalizing the MLE machinery of Session 5) and search the DAG space greedily (add/remove/reverse one edge at a time) rather than exhaustively.

- **Constraint-based search**: run a sequence of conditional-independence tests (the sample analogue of Session 1's Definition 1.2) to infer which edges *must* be absent, then orient the remaining edges using the collider-detection logic from Session 2 (a v-structure $X \to Z \leftarrow Y$ is distinguishable from a chain precisely because X and Y test as independent while X, Y given Z do not).

In practice, on real applied problems, structure is very often *not* learned purely from data at all: an expert-elicited graph (like our student network or a clinical DAG built from established medical knowledge) is preferred whenever available, both because data-driven structure search is statistically unreliable at modest sample sizes and — as Session 7 will make sharply clear — because a graph learned purely from *observational, correlational* structure can be silent or actively misleading about which edges represent genuine causal mechanisms versus mere statistical association. We will not resolve that tension today; it is the very question the second half of this course is devoted to.

20 When Exact Inference Fails: The Case for Sampling

Recall Session 3's headline result: VE costs $O(n \cdot d^{w+1})$ where w is the treewidth. Many real graphical models — densely connected social networks, image grids with strong neighbor coupling, gene regulatory networks — have treewidth comparable to n itself, making exact inference computationally hopeless regardless of how cleverly we order eliminations. *Sampling-based approximate inference* trades exactness for tractability: instead of computing $P(y \mid e)$ exactly, we draw samples from (a distribution close to) P, and estimate $P(y \mid e)$ by counting.

20.1 Forward (Ancestral) Sampling

Definition 20.1 (Ancestral sampling)**.** Given a Bayesian network with no evidence, sample every variable in an order consistent with the DAG (parents before children, always possible since the graph is acyclic): $x_i \sim P(X_i \mid \mathrm{pa}(x_i))$, using already-sampled parent values.

This produces exact i.i.d. samples from the joint $P(\mathbf{x})$ with no approximation at all — it is simply the chain rule (Session 2) turned into a generative procedure. The catch appears once we need to condition on evidence: if evidence $E = e$ is rare, most sampled trajectories will simply not match e and must be discarded, wasting nearly all the computation (*rejection sampling*). *Importance sampling* improves on this by sampling from a more convenient proposal distribution Q and reweighting each sample by $P(\mathbf{x})/Q(\mathbf{x})$ rather than discarding mismatches outright — a technique we will meet again, doing essentially the same reweighting job, as inverse propensity weighting in Session 10.

20.2 Markov Chain Monte Carlo: Gibbs Sampling

When even constructing a good proposal Q is hard (common for densely-connected Markov networks, where there is no natural "ancestral" order at all), we turn to Markov chain Monte Carlo (MCMC): build a Markov chain over the state space whose *stationary distribution* is exactly the target P, then run it long enough and treat the states visited as (correlated, but eventually representative) samples from P.

Algorithm 3 Gibbs Sampling

Initialize $\mathbf{x}^{(0)}$ arbitrarily.
for $t = 0, 1, 2, \ldots$ **do**
  for each variable X_i (in any order, or at random) **do**
    Sample $x_i^{(t+1)} \sim P\big(X_i \mid x_{-i}^{(t)}\big)$, the full conditional given all other current values.
  end for
end for

The full conditional $P(X_i \mid x_{-i})$ has a beautiful simplification in a Markov network (Session 4): it only depends on X_i's *neighbors* in the graph, since every other variable is d-... (rather, Markov-)blanketed away by the neighbors' potentials — another direct payoff of the undirected graphical structure.

Proposition 20.2 (Gibbs sampling is correct, informally)**.** *The Markov chain defined by Gibbs sampling has the target Gibbs distribution P as its unique stationary distribution: if $\mathbf{x}^{(t)} \sim P$, then $\mathbf{x}^{(t+1)} \sim P$ as well.*

The proof (appendix) is a direct check of *detailed balance*: the chain is exactly as likely to move from state a to state b as from b to a, weighted by P, which is a sufficient condition for P to be stationary. In practice, the chain needs some number of iterations ("burn-in") before it forgets its (usually poor) initialization and starts producing draws representative of P; Figure 17 shows this concretely on a small worked example below.

Intuition

Do not confuse the number of Gibbs sweeps with the number of independent samples: consecutive draws are correlated (each differs from the last by only one variable's resampling), so practitioners typically discard a burn-in period *and* thin the chain (keep only every k-th sample) before treating the output as approximately i.i.d. draws from P.

21 Worked Examples

Worked Problem 21.1 (Gibbs sampling a small Markov network, exactly checked)**.** Consider three binary spins $X_1, X_2, X_3 \in \{-1, +1\}$ arranged in a triangle (every pair connected), with pairwise potentials $\phi(x_i, x_j) = \exp(J\, x_i x_j)$, $J = 0.9$, favoring agreement (the same style of model as Session 4's grid, small enough here that we can also compute the exact answer by brute-force enumeration over all $2^3 = 8$ states, for comparison). We ran a real Gibbs sampler for 3000 sweeps, discarding the first 50 as burn-in (Figure 17); the two all-agreeing configurations $(+,+,+)$ and $(-,-,-)$ dominate the true distribution (each carries probability 0.462, versus 0.013 for each of the six disagreeing configurations), and the sampler's empirical frequencies after burn-in closely match these exact values (Figure 18), confirming Proposition 20.2 numerically without ever computing the partition function Z directly — exactly the situation we will face in larger, intractable models where brute-force checking is no longer available.

Worked Problem 21.2 (Estimating a rare-evidence query by ancestral sampling)**.** Suppose in the student network $P(L = \text{strong})$ is rare (say true probability 0.02). To estimate $P(I = \text{high} \mid L = \text{strong})$ by naive rejection sampling from ancestral draws, we would expect to need roughly

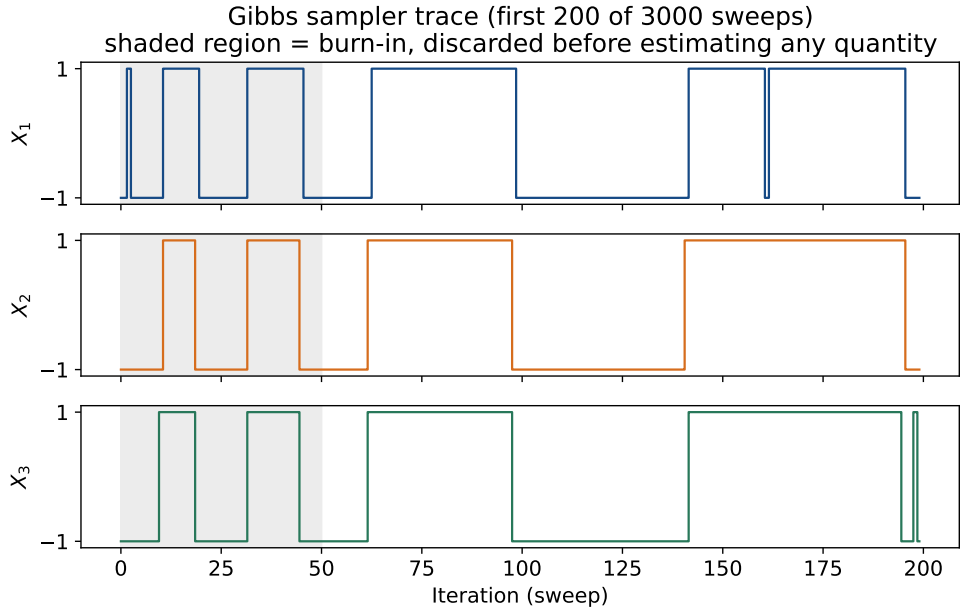


Figure 17: Trace of each spin variable across the first 200 Gibbs sweeps. The shaded region is discarded as burn-in.

$1/0.02 = 50$ full samples on average just to see *one* usable draw matching the evidence — for a query on a rarer event (say probability 0.0002), that becomes 5,000 wasted samples per usable one. This is exactly the motivation for importance sampling (reweight instead of reject) and, in harder cases with no convenient proposal, for MCMC methods like Gibbs sampling that never need to "match" evidence by chance — evidence is instead fixed once and the sampler only ever resamples the unobserved variables' full conditionals.

Worked Problem 21.3 (Structure learning sanity check)**.** Given complete data on the student network's five variables, a constraint-based structure learner tests marginal independence between every pair. It should find $D \perp\!\!\!\perp I$ (no common cause assumed, no edge), and — critically, per Session 2's Exercise 2.2 — it should find D and S marginally independent *but* dependent given G. A correct constraint-based algorithm uses exactly this asymmetric pattern to infer that G must be a collider with parents D and I (rather than, say, a chain), recovering the correct edge orientations from data alone — but notice that this recovers only the graph's *independence skeleton and colliders*; whether an edge like $D \to G$ represents difficulty genuinely causing grade difficulty, versus a purely associational convention, is a question this algorithm has no way to answer. That distinction is exactly what Session 7 forces us to confront.

Additional Worked Examples (Instructor Supplement)

Worked Problem 21.4 (A Gibbs full conditional, computed exactly)**.** For the triangle spin model ($\phi(x_i, x_j) = e^{J x_i x_j}$, J=0.9, spins in $\{-1, +1\}$), compute the full conditional $P(X_1 \mid X_2{=}{+1}, X_3{=}{+1})$ used inside every sweep of Gibbs sampling.

Solution. X_1's only neighbors are X_2, X_3 (its Markov blanket in this graph), so $P(X_1 \mid X_2, X_3) \propto \phi(X_1, X_2)\phi(X_1, X_3)$. For $X_1{=}{+1}$: $e^{0.9(1)(1)} e^{0.9(1)(1)} = e^{1.8} \approx 6.0496$. For $X_1{=}{-1}$:

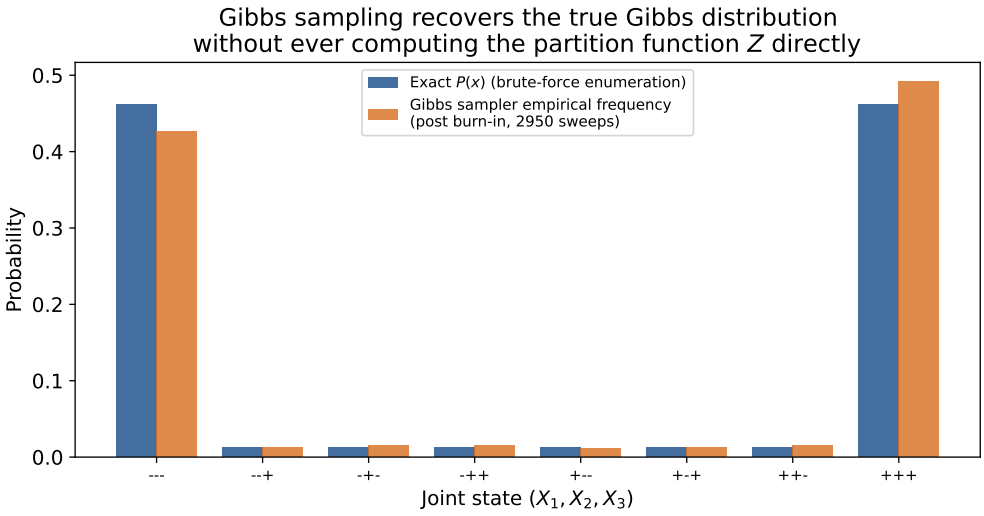


Figure 18: Exact probabilities (brute-force enumeration) versus Gibbs sampler empirical frequencies, post burn-in. The two agree closely.

$e^{0.9(-1)(1)}e^{0.9(-1)(1)} = e^{-1.8} \approx 0.1653$. Normalizing over the total $6.0496 + 0.1653 = 6.2149$:

$$P(X_1{=}{+}1 \mid X_2{=}1, X_3{=}1) = \frac{6.0496}{6.2149} \approx \mathbf{0.9734}, \qquad P(X_1{=}{-}1 \mid \cdots) \approx \mathbf{0.0266}.$$

This confirms, with real numbers, the claim that the full conditional depends only on X_1's neighbors and requires normalizing over only the 2 values of X_1 itself — never the full $2^3 = 8$-state joint — which is exactly what makes Gibbs sampling cheap per step even when the joint distribution (and its partition function Z) is intractable to compute directly.

Worked Problem 21.5 (Rejection sampling can fail even with "enough" draws on average)**.** If rare evidence has true probability $p = 0.02$, the expected number of ancestral draws needed for one accepted sample is $1/p = 50$ (as noted above). Suppose we budget exactly 100 draws (twice the expectation). What is the probability we get *zero* usable samples?

Solution. Each draw independently matches the evidence with probability $p = 0.02$, so the number of matches in 100 draws is Binomial$(100, 0.02)$, and

$$P(\text{zero matches in 100 draws}) = (1 - 0.02)^{100} = 0.98^{100}.$$

Using $\ln(0.98) \approx -0.020203$, we get $100 \ln(0.98) \approx -2.0203$, so $0.98^{100} \approx e^{-2.0203} \approx \mathbf{0.133}$. Even at *twice* the expected number of draws, there is still roughly a 13% chance of walking away with no usable sample at all — a stark, quantitative reason rejection sampling is considered impractical for rare evidence, well beyond the qualitative "you'll waste a lot of samples" framing, and a concrete motivation for both importance sampling (reweight instead of discard) and MCMC (never needs to match evidence by chance in the first place).

Worked Problem 21.6 (A BIC calculation, structure search made concrete)**.** A structure-learning algorithm compares two candidate structures for (A, B) using $N = 100$ fully-observed records: 60 have $A{=}1$, 40 have $A{=}0$; among $A{=}1$, $B{=}1$ occurs 48 times; among $A{=}0$, $B{=}1$ occurs 8 times. Compare BIC $= \ell(\hat{\theta}) - \frac{k}{2} \log N$ for the edge model $(A \to B)$ versus the no-edge model $(A \perp B)$.

Solution. **Edge model.** MLEs: $\hat{P}(A{=}1) = 0.6$, $\hat{P}(B{=}1 \mid A{=}1) = 48/60 = 0.8$, $\hat{P}(B{=}1 \mid A{=}0) = 8/40 = 0.2$. Log-likelihood: $\ell_{\text{edge}} = 60 \ln 0.6 + 40 \ln 0.4 + 48 \ln 0.8 + 12 \ln 0.2 + 8 \ln 0.2 +$

$32\ln 0.8$. Grouping the B-terms (80 occurrences of $\ln 0.8$, 20 of $\ln 0.2$): $\ell_{\text{edge}} \approx 60(-0.5108) + 40(-0.9163) + 80(-0.2231) + 20(-1.6094) \approx -30.65 - 36.65 - 17.85 - 32.19 = \mathbf{-117.34}$. With $k_{\text{edge}} = 3$ free parameters (one for $P(A)$, two for $P(B \mid A)$), $\text{BIC}_{\text{edge}} \approx -117.34 - 1.5\ln(100) \approx -117.34 - 6.91 = \mathbf{-124.25}$.

No-edge model. $\hat{P}(B{=}1) = 56/100 = 0.56$ (pooled). $\ell_{\text{indep}} = [60\ln 0.6 + 40\ln 0.4] + [56\ln 0.56 + 44\ln 0.44] \approx -67.30 + (-68.57) = \mathbf{-135.87}$. With $k_{\text{indep}} = 2$, $\text{BIC}_{\text{indep}} \approx -135.87 - 1(4.605) = \mathbf{-140.48}$.

Since $-124.25 > -140.48$ by a wide margin (≈ 16.2), the edge model wins decisively: the data's strong dependence (0.8 vs. 0.2) far outweighs the extra parameter's complexity penalty at $N = 100$. This is exactly the score-based search of the main text, made concrete enough to hand-verify rather than taken on faith.

Appendix: DAG-Counting Recurrence and Detailed Balance

Robinson's recurrence. The number $a(n)$ of DAGs on n labeled nodes satisfies $a(0) = 1$ and

$$a(n) = \sum_{k=1}^{n} (-1)^{k+1} \binom{n}{k} 2^{k(n-k)} a(n-k),$$

obtained by an inclusion-exclusion argument over the possible sets of "source" nodes (nodes with no incoming edges) at each recursive level.

Detailed balance for Gibbs sampling. Let $T(\mathbf{x} \to \mathbf{x}')$ be the Gibbs transition probability for resampling coordinate i (all other coordinates unchanged, $x_{-i} = x'_{-i}$). Detailed balance requires $P(\mathbf{x})T(\mathbf{x} \to \mathbf{x}') = P(\mathbf{x}')T(\mathbf{x}' \to \mathbf{x})$. Since $T(\mathbf{x} \to \mathbf{x}') = P(x'_i \mid x_{-i})$ by construction, and using $P(\mathbf{x}) = P(x_i \mid x_{-i})P(x_{-i})$,

$$P(\mathbf{x})T(\mathbf{x} \to \mathbf{x}') = P(x_i \mid x_{-i})P(x_{-i})\,P(x'_i \mid x_{-i}) = P(x_{-i})P(x_i \mid x_{-i})P(x'_i \mid x_{-i}),$$

which is symmetric in $x_i \leftrightarrow x'_i$ (using $x_{-i} = x'_{-i}$), so it equals $P(\mathbf{x}')T(\mathbf{x}' \to \mathbf{x})$ by the same computation with roles swapped. Detailed balance implies P is a stationary distribution of the chain (summing the balance equation over $\mathbf{x}$ gives $\sum_{\mathbf{x}} P(\mathbf{x})T(\mathbf{x} \to \mathbf{x}') = P(\mathbf{x}')$, the definition of stationarity); ergodicity (needed for the chain to actually *converge* to this stationary distribution regardless of start) requires positivity of all full conditionals, which holds automatically whenever the Gibbs potentials are everywhere strictly positive.

Connections and Takeaways

We now have a complete Act I toolkit: represent independence with a graph (Sessions 2, 4), compute exactly when structure allows it (Session 3), learn parameters and handle latent variables (Session 5), and fall back to sampling — including MCMC — whenever exact computation is out of reach (today). Every one of these tools computes properties of a single, fixed joint distribution P obtained by *observing* the world.

Here is the question none of Sessions 1–6 can answer: if I *change* the world — give every patient the drug rather than merely observing who happened to take it, force an ad in front of every customer rather than observing who happened to see it — what happens to Y? This is not a question about $P(Y \mid X = x)$, and, as **Session 7** will demonstrate with real

numbers, answering it as if it were can be actively wrong, sometimes even reversing the sign of the true effect. That demonstration is the hinge of the entire course.

Session 7: Correlation vs. Causation: The Conceptual Pivot

Opening Motivation

Every tool built in Sessions 1–6 — CI, DAGs, variable elimination, Markov networks, EM, MCMC — computes properties of a single fixed probability distribution P, generated by observing the world as it already is. Today we show, with real published data, that this entire toolkit can give you the *wrong sign* of an effect if you ask it the wrong question. Not a slightly-off estimate: the wrong sign, on a scale that has cost real patients real outcomes. This is not a footnote or a caveat to append to Sessions 1–6; it is the reason a second, distinct mathematical formalism (Sessions 8–10) is necessary at all.

22 Two Real Reversals

Example 22.1 (UC Berkeley graduate admissions, 1973)**.** In fall 1973, UC Berkeley's graduate admissions showed a troubling aggregate pattern: of 8442 male applicants, 44.5% were admitted, versus only 30.4% of 4321 female applicants (Bickel, Hammel & O'Connell, *Science*, 1975) — see Figure 19, left panel. This gap looks like exactly the kind of evidence a sex-discrimination lawsuit would want. But when the university's own statisticians broke the same data down *by department* (right panel, six largest departments), something remarkable appeared: in four of the six departments, women were admitted at an equal or *higher* rate than men. Department A alone admitted 82.4% of its (relatively few) female applicants versus 62.1% of men.

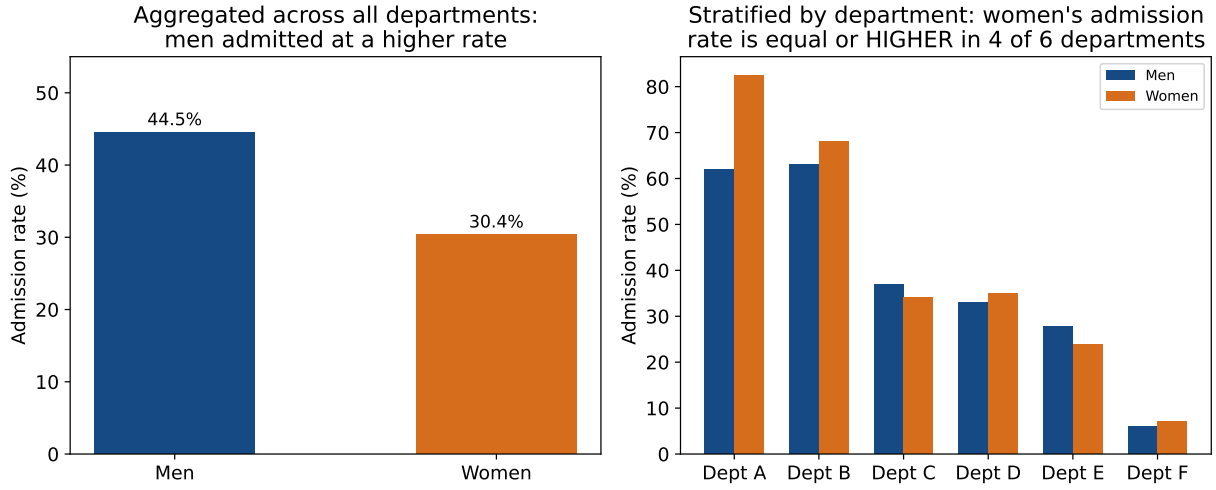


Figure 19: Aggregated (left) vs. department-stratified (right) admission rates, UC Berkeley 1973. Real data (Bickel, Hammel & O'Connell, 1975).

The resolution: women disproportionately applied to competitive departments with low admission rates for everyone (e.g., Department C admits under 37% of anyone), while men disproportionately applied to less competitive departments (Department A admits over 60% of anyone). Department choice acted as a *confounder* — a common cause of both "which group you're com-

pared within" and "your chance of being admitted" — and aggregating across it mixed the effect of department choice into what looked like a gender effect.

Example 22.2 (Kidney stone treatment, Charig et al. 1986)**.** A comparative study of two kidney-stone treatments — open surgery (Treatment A) and percutaneous nephrolithotomy (Treatment B) — reported, in aggregate, that Treatment B succeeded 82.6% of the time versus 78.0% for Treatment A (Figure 20, left), suggesting B is the better treatment. Yet stratified by stone size (Figure 20, right, using the standard tabulation by Julious & Mullee, *BMJ*, 1994): for *small* stones, A succeeds 93.1% of the time versus B's 86.7%; for *large* stones, A succeeds 73.0% versus B's 68.8%. Treatment A is **better in both subgroups**, yet worse overall. This is not a rounding artifact — it is a full reversal of direction.

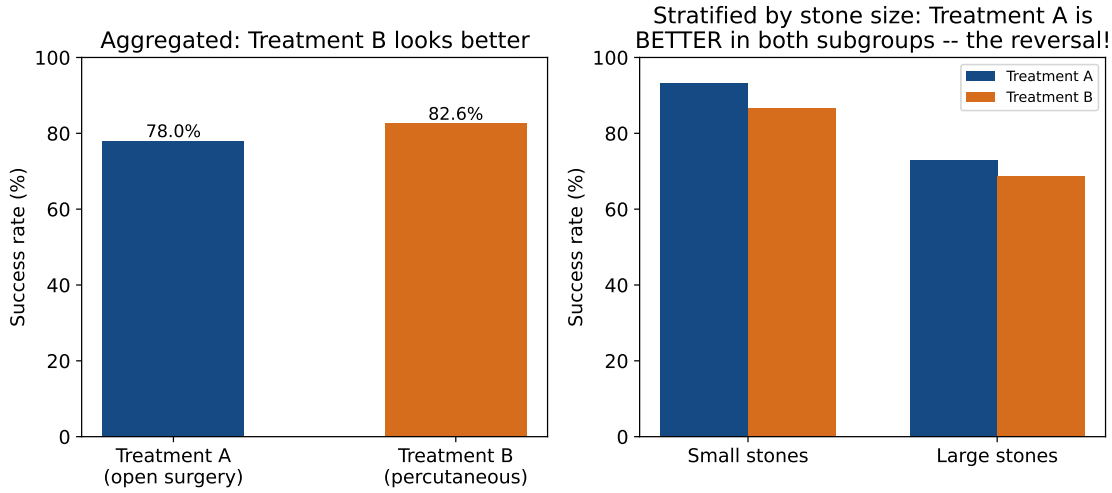


Figure 20: Aggregated (left) vs. severity-stratified (right) treatment success rates. Real data (Charig et al., 1986).

The mechanism is identical to Berkeley's: physicians preferentially used the less invasive Treatment B on *small* (easier) stones and reserved open surgery (Treatment A) for *large*, harder cases. Stone size drives both which treatment a patient received and how likely they were to recover regardless of treatment. Aggregating over stone size mixes "how severe was your case" into what looks like a treatment effect.

Definition 22.3 (Simpson's paradox)**.** An association between X and Y that holds in the aggregated population reverses (or vanishes) when the population is stratified by a third variable Z.

23 The Graphical Diagnosis

Both examples share one causal structure (Figure 21): a variable Z (department choice; stone severity) is a common cause of both X (gender-driven applicant pool composition; treatment choice) and Y (admission; recovery). In Session 2's vocabulary, Z is a *fork*: $X \leftarrow Z \rightarrow Y$. Marginally, this fork *induces* an association between X and Y that has nothing to do with any direct effect of X on Y — exactly the enthusiast example from Session 1 and the "explaining away" logic from Session 2, but now with the arrow of practical consequence reversed: instead of a spurious *independence*

being broken, we have a spurious *dependence* being created between the treatment we care about and the outcome we care about.

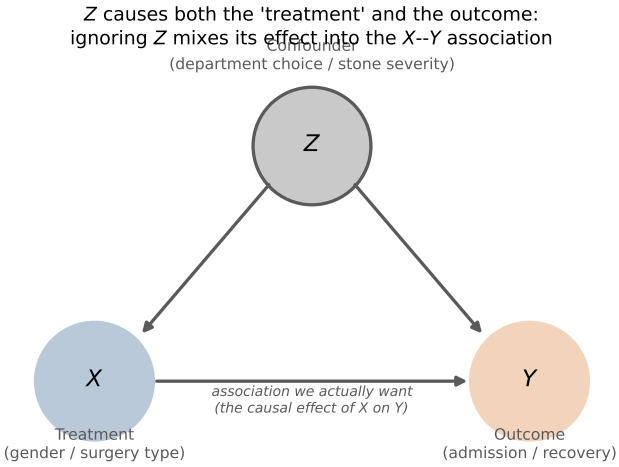


Figure 21: The shared causal structure behind both reversals: a confounder Z drives both "treatment" and outcome.

Definition 23.1 (Confounder, informally)**.** A variable Z is a *confounder* for the relationship between X and Y if Z causally influences both X and Y. Naively comparing $P(Y \mid X = 1)$ to $P(Y \mid X = 0)$ then reflects a mixture of (i) any genuine effect of X on Y and (ii) the spurious association induced by Z.

The seemingly obvious fix — "stratify by (control for) Z" — is exactly what resolved both examples above, and it will be formalized properly as the *backdoor criterion* in Session 9. But notice we have already seen, in Session 2's Exercise 2.2, that conditioning is not *always* the right move: conditioning on a **collider** ($X \to Z \leftarrow Y$, rather than a confounding fork $X \leftarrow Z \to Y$) *creates* spurious association where none existed before. A blanket rule of "control for everything you measured" is therefore not merely inelegant — it can be actively wrong, and can manufacture exactly the kind of paradox we've just resolved, in the opposite direction. This tension — when to condition, when not to — is precisely why Session 9 will insist on a checkable graphical *criterion* rather than a rule of thumb.

24 Why $P(Y \mid X{=}x)$ Is the Wrong Quantity

Here is the sharpest way to state what went wrong in both examples. The number we computed, $P(Y = 1 \mid X = x)$, answers: *"among the units we observed that happened to have $X = x$, what fraction had $Y = 1$?"* It is a fact about who ended up in each group — and in both examples, who ended up in each group was itself driven by Z. What we actually want to know is different: *"if we were to take a patient (or applicant) and force X to equal x, overriding whatever process would ordinarily determine X for them, what would happen to Y?"* This is not a question about a subpopulation defined by an observed value of X; it is a question about an *action* performed on a unit, irrespective of how that unit would ordinarily have arrived at that value of X.

We introduce notation for this now, informally, and formalize it fully next session:

$P(y \mid x)$ (observational: "who has $X = x$?") vs. $P\big(y \mid \mathrm{do}(x)\big)$ (interventional: "set $X = x$ for everyone"

Key Takeaway 24.1. In general, $P(y \mid x) \neq P(y \mid \text{do}(x))$. Every session so far has computed the left-hand side. Nothing in Sessions 1–6 — no amount of more data, better inference algorithms, or more careful sampling — closes this gap, because the gap is not a statistical estimation problem. It is a conceptual one: the DAGs of Sessions 2 and 4 encoded only a factorization of P, and a factorization alone does not specify what happens when a piece of the generative process is overridden rather than observed.

25 Worked Examples

Worked Problem 25.1 (Reconstructing the Berkeley reversal arithmetically)**.** Using only Department A (825 men, 512 admitted; 108 women, 89 admitted) and Department C (325 men, 120 admitted; 593 women, 202 admitted), verify that women's aggregate admission rate can fall below men's even though women's rate is higher in Department A specifically.

Solution. In Department A alone: men $512/825 = 62.1\%$, women $89/108 = 82.4\%$ — women higher. In Department C alone: men $120/325 = 36.9\%$, women $202/593 = 34.1\%$ — men slightly higher, but the more important asymmetry is where the applicants concentrate: department C received far more of its total applicant pool from women (593) than men (325), and C is intrinsically a lower-admission-rate department for everyone. Aggregating A and C together: men $(512 + 120)/(825 + 325) = 632/1150 = 55.0\%$, women $(89 + 202)/(108 + 593) = 291/701 = 41.5\%$ — the aggregate now favors men even though women did at least as well in the higher-admitting department, purely because women's applications concentrated more heavily in the lower-admitting one. This is Simpson's paradox reduced to two departments and pure arithmetic: no discrimination coefficient is needed to produce the gap, only a differential rate of applying to harder departments.

Worked Problem 25.2 (A marketing analogue)**.** An online retailer notices that customers who were shown a promotional ad convert (make a purchase) 18% of the time, versus 9% for those not shown the ad – seemingly a strong case for showing the ad to everyone. But the ad-targeting algorithm preferentially shows ads to customers with high prior purchase intent (say, based on recent browsing activity), and high-intent customers convert at a high rate *regardless* of the ad. If, hypothetically, high-intent customers convert at 25% whether or not they see the ad, and low-intent customers convert at 5% whether or not they see the ad, but 70% of ad-shown customers are high-intent versus only 20% of not-shown customers, then the naive comparison recovers almost exactly the "effect" implied above ($0.7(0.25) + 0.3(0.05) = 0.19$ vs. $0.2(0.25) + 0.8(0.05) = 0.09$) with *zero true causal effect of the ad at all* — purchase intent is a confounder exactly as department was for Berkeley and stone size was for kidney stones. This is the vignette we will resolve properly with a formal identification argument in Sessions 9 and 10.

Worked Problem 25.3 (Distinguishing confounding from mere sampling noise)**.** A skeptic argues both reversals above might just be small-sample flukes. Verify this is not so for Berkeley: the overall sample sizes (8442 and 4321) are large enough that a 14-percentage-point gap is not plausibly due to chance, and the same is true within each department once we account for the actual counts involved. The point of stratification is not to *explain away* a real effect as noise, but to reveal that the aggregated number was measuring a mixture of two different things (department-level admission difficulty and gender) that happened to move together in the data — a structural fact about how the applicant pool was formed, not a statistical fluke.

Additional Worked Examples (Instructor Supplement)

Worked Problem 25.4 (A third real reversal: batting averages)**.** Simpson's paradox is not confined to admissions or medicine. In the 1995 and 1996 Major League Baseball seasons, David Justice and Derek Jeter's batting averages (hits/at-bats) were:

$$\text{1995: Justice } \tfrac{104}{411} \approx .253,\ \text{Jeter } \tfrac{12}{48} = .250; \qquad \text{1996: Justice } \tfrac{45}{140} \approx .321,\ \text{Jeter } \tfrac{183}{582} \approx .314.$$

Justice has the higher average in *both* individual seasons. Yet combined over the two years,

$$\text{Justice: } \frac{104+45}{411+140} = \frac{149}{551} \approx \mathbf{.270}, \qquad \text{Jeter: } \frac{12+183}{48+582} = \frac{195}{630} \approx \mathbf{.310},$$

and Jeter's combined average is *higher*. The mechanism is identical to Berkeley and the kidney stones: year is a confounder, since Jeter had far more at-bats in 1996 (the higher-average year for both players, .321 and .314, versus the lower-average 1995), so his combined average is weighted more heavily toward his better year, while Justice's at-bats were split more evenly. No baseball skill differential is needed to produce this reversal — only a differential in *how many at-bats occurred in which year*, exactly mirroring the differential application rates to departments (Berkeley) and differential treatment assignment by stone size (kidney stones).

Worked Problem 25.5 (Simpson's paradox in a continuous outcome, not just a rate)**.** (Hypothetical, for illustration.) Two companies' employees report job-satisfaction scores by seniority. Company X: 90 senior staff, average satisfaction 9.0; 10 junior staff, average satisfaction 5.0. Company Y: 10 senior staff, average satisfaction 9.5; 90 junior staff, average satisfaction 5.5. Within *each* seniority level, Company Y does better (junior: $5.5 > 5.0$; senior: $9.5 > 9.0$). Yet aggregated:

$$\text{Company X: } \frac{90(9.0)+10(5.0)}{100} = \frac{810+50}{100} = \mathbf{8.60}, \qquad \text{Company Y: } \frac{10(9.5)+90(5.5)}{100} = \frac{95+495}{100} = \mathbf{5.90}.$$

Company X's aggregate score (8.60) is far *higher* than Company Y's (5.90), reversing Y's within-group advantage in both strata. This confirms Simpson's paradox is a property of any weighted average, not merely proportions: Company X simply has a much more senior (and hence, within-level, happier) workforce composition, letting a systematically *worse* performer at every seniority level post the better headline number — exactly the same weighted-average arithmetic behind Session 7's admissions and treatment examples, here applied to a continuous score rather than a binary success rate.

Connections and Takeaways

This session had no new theorem to state and none to defer to an appendix, deliberately: the point was to make the insufficiency of $P(Y \mid X = x)$ felt, with real numbers, before we spend three sessions building the machinery to fix it. Three things carry forward. First, the vocabulary: confounder (a fork creating spurious X–Y association) versus collider (whose naive conditioning creates the opposite mistake) — both graphical concepts already available since Session 2, now understood as double-edged. Second, the diagnosis: $P(y \mid x) \neq P(y \mid \text{do}(x))$ in general, and no amount of more data or better inference algorithms closes this gap by itself. Third, and most important: the missing ingredient is not a new independence-reading rule, but a new kind of arrow — one that means "generates" rather than merely "factorizes with." **Session 8** builds exactly that: the structural causal model, in which the DAGs we have been drawing since Session 2 are reinterpreted as literal generative mechanisms, robust

enough to support the question "what if this piece were overridden?" that observation alone can never answer.

Session 8: Structural Causal Models

Opening Motivation

Session 7 ended on a gap: $P(y \mid x) \neq P(y \mid \mathrm{do}(x))$ in general, and nothing in our toolkit so far can compute the right-hand side, because a DAG with CPTs only ever specified a *factorization* of P — it never said what happens if a piece of that factorization is overridden rather than observed. Today we fix this by strengthening what an arrow means. Instead of "X_i's distribution depends on its parents," a structural causal model (SCM) says "X_i is *computed from* its parents by a fixed mechanism" — a claim strong enough to survive having one mechanism swapped out, which is exactly what an intervention does.

26 Structural Causal Models

Definition 26.1 (Structural causal model)**.** A structural causal model $M = (\mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}))$ consists of:

- *exogenous* variables $\mathbf{U}$ — background factors external to the model, with a specified joint distribution $P(\mathbf{U})$;
- *endogenous* variables $\mathbf{V} = \{V_1, \ldots, V_n\}$ — the variables of interest, determined within the model;
- a set of *structural equations* $\mathcal{F} = \{f_1, \ldots, f_n\}$, one per endogenous variable, of the form

$$V_i := f_i\big(\mathrm{pa}(V_i), U_i\big),$$

  where $\mathrm{pa}(V_i) \subseteq \mathbf{V} \setminus \{V_i\}$ and $U_i \subseteq \mathbf{U}$ are the direct causes of V_i.

The symbol $:=$ (assignment, not equality) is deliberate: $V_i := f_i(\cdots)$ says V_i is *computed* by this mechanism, not merely that this equation happens to hold. This asymmetry — present in the notation but invisible in an ordinary equals sign — is precisely what will let us "delete" one equation while keeping the rest, which an algebraic equality could never justify (you cannot selectively erase one equation from a system of equalities without changing what the whole system claims).

Every SCM induces an ordinary DAG G: draw an edge $V_j \to V_i$ whenever $V_j \in \mathrm{pa}(V_i)$. And crucially:

Proposition 26.2 (Consistency with Sessions 2 and 4)**.** *The distribution $P(\mathbf{v})$ induced by an SCM (obtained by pushing $P(\mathbf{U})$ forward through $\mathcal{F}$) factorizes exactly as $\prod_i P(v_i \mid \mathrm{pa}(v_i))$ over the induced DAG G — the same Bayesian network factorization from Session 2.*

So nothing from Sessions 1–6 is discarded: an SCM is a *refinement* of a Bayesian network, adding mechanism on top of factorization, and every observational computation (VE, MLE, EM, sampling) still applies unchanged to the induced $P(\mathbf{v})$. What's new is that the SCM also licenses questions the factorization alone could not answer.

Example 26.3 (A linear SCM for wages)**.** Let A (ability), X (years of education), Y (income) be linked by

$$A := U_A, \qquad X := 2A + U_X, \qquad Y := 3X + 5A + U_Y,$$

with U_A, U_X, U_Y mutually independent standard normal noises (Figure 22). This is exactly the confounding structure of Session 7: A is a common cause of X and Y (a fork), so observing X carries information about A, which in turn carries information about Y through the direct arrow $A \to Y$ — inflating any naive association between X and Y beyond the true structural coefficient 3.

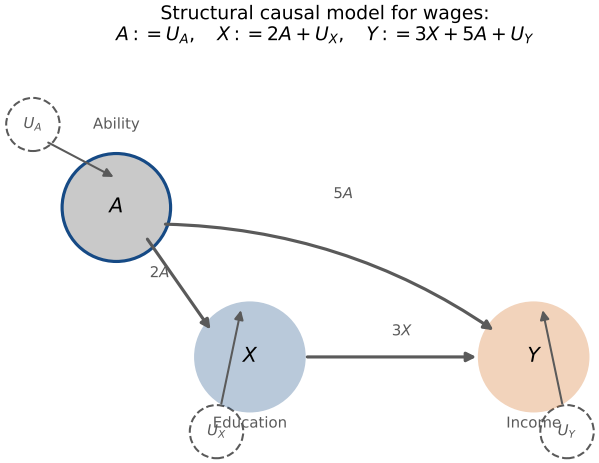


Figure 22: An SCM as a generative mechanism: solid nodes are endogenous, dashed nodes are exogenous noise.

27 Intervention as Graph Surgery

Definition 27.1 (Intervention)**.** The intervention $\mathrm{do}(X = x)$ on SCM M produces a new (*mutilated*, or *submodel*) SCM M_x by replacing the structural equation for X with the constant assignment $X := x$, deleting X's dependence on $\mathrm{pa}(X)$ and U_X, and leaving every other equation in $\mathcal{F}$ unchanged. The interventional distribution $P(\mathbf{y} \mid \mathrm{do}(x))$ is the distribution of $\mathbf{Y}$ induced by M_x.

Graphically (Figure 23), this deletes every incoming edge to X while keeping all outgoing edges intact — X still causes whatever it caused before, it is simply no longer caused by anything itself.

Theorem 27.2 (Truncated factorization)**.** *For an intervention* $\mathrm{do}(X = x)$ *on variables* $\mathbf{V} = \{V_1, \ldots, V_n\}$*,*

$$P(v_1, \ldots, v_n \mid \mathrm{do}(x)) = \begin{cases} \displaystyle\prod_{i:\, V_i \neq X} P\big(v_i \mid \mathrm{pa}(v_i)\big) & \text{if } x \text{ is consistent with the assignment,} \\ 0 & \text{otherwise,} \end{cases}$$

i.e., simply drop the factor $P(x \mid \mathrm{pa}(x))$ *from the ordinary chain-rule factorization (Session 2) and fix* $X = x$ *everywhere else.*

This is the single most important formula of this session, and it is the bridge connecting the entire directed-graphical-model machinery of Sessions 2–3 directly to causal queries: computing an interventional distribution is *exactly* an ordinary Bayesian-network computation (Session 3's

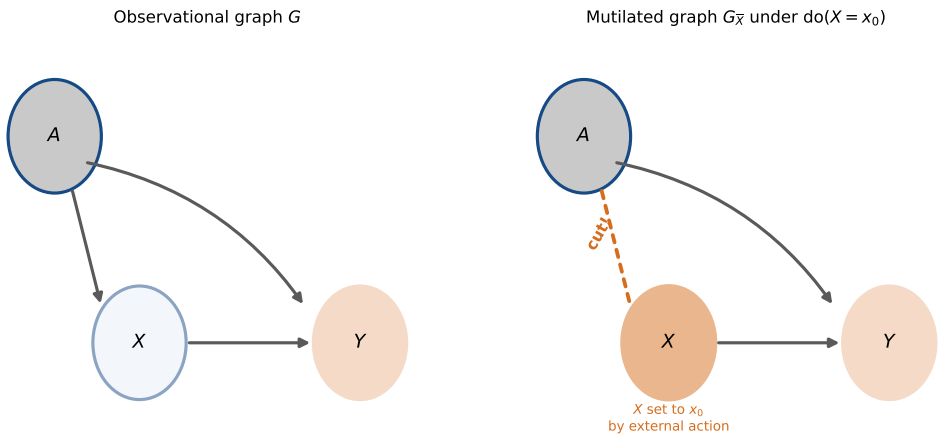


Figure 23: Graph surgery: $\text{do}(X = x_0)$ deletes X's incoming edges (here, from confounder A) while keeping its outgoing edges to Y.

variable elimination applies immediately) on the *mutilated* graph, using the very same CPTs as the original network for every equation except X's. No new algorithm is needed — only a different graph.

Intuition

Contrast conditioning with intervening using our confounding example. Conditioning on $X = x$ (Session 7's $P(y \mid x)$) restricts attention to a subpopulation — everyone in that subpopulation still got their X value through the ordinary mechanism $X := 2A + U_X$, so if X is high partly because A is high, that information about A leaks into what we know about Y. Intervening, $\text{do}(X = x)$, severs this: every unit is forced to $X = x$ regardless of its A, so the leak is cut at the source, exactly as Theorem 27.2 formalizes by simply deleting X's factor.

28 Counterfactuals: A Glimpse

Interventions ask "what would happen to a population if we set $X = x$?" Counterfactuals ask a sharper, unit-specific question: "for this specific unit, who we observed with $X = x$ and $Y = y$, what would Y have been had X been x' instead?" SCMs answer this via a three-step procedure:

1. **Abduction**: given the observed evidence, update the distribution of the exogenous noise $\mathbf{U}$ (for a specific unit with fully observed variables, this pins $\mathbf{U}$ down exactly, since the structural equations are invertible given all endogenous values).
2. **Action**: perform $\text{do}(X = x')$ (graph surgery, Definition 27.1) on the model.
3. **Prediction**: compute Y in this modified model, using the *same* (abduced) value of $\mathbf{U}$ as this specific unit.

Remark 28.1 (Bridge to potential outcomes). Students arriving from the potential-outcomes (Neyman–Rubin) tradition will recognize $Y_x(u)$ — "the value Y would take for unit u if X were set to x" — as exactly the SCM counterfactual just defined, usually written $Y(x)$ or Y_x. The two frameworks

are provably equivalent in what they can express; SCMs simply make the generative mechanism explicit, which is convenient for reading identifiability directly off a graph (Session 9), while potential outcomes are often more convenient for stating assumptions directly in terms of estimands (Session 10). We will use both notations as convenient from here on.

29 Worked Examples

Worked Problem 29.1 (The wage SCM: observational vs. interventional slope)**.** Using Example 26.3, compute (i) the slope of the observational regression of Y on X, and (ii) the true causal effect $\frac{d}{dx}\mathbb{E}[Y \mid \mathrm{do}(X = x)]$.

Solution. Substituting $X = 2A + U_X$ into $Y = 3X + 5A + U_Y$ gives $Y = 6A + 3U_X + 5A + U_Y = 11A + 3U_X + U_Y$. With unit-variance independent noises, $\mathrm{Cov}(X, Y) = \mathrm{Cov}(2A + U_X,\ 11A + 3U_X + U_Y) = 22\,\mathrm{Var}(A) + 3\,\mathrm{Var}(U_X) = 22 + 3 = 25$, and $\mathrm{Var}(X) = 4\,\mathrm{Var}(A) + \mathrm{Var}(U_X) = 4 + 1 = 5$, giving observational slope $25/5 = 5$. For the interventional quantity, Theorem 27.2 says: drop X's equation, fix $X = x$, keep $A := U_A$ and $Y := 3X + 5A + U_Y$ unchanged; then $\mathbb{E}[Y \mid \mathrm{do}(x)] = 3x + 5\mathbb{E}[A] + \mathbb{E}[U_Y] = 3x$ (since $\mathbb{E}[A] = \mathbb{E}[U_Y] = 0$), so $\frac{d}{dx}\mathbb{E}[Y \mid \mathrm{do}(x)] = 3$ exactly — matching the structural coefficient by construction, and matching Session 7's warning precisely: the naive association (5) overstates the true causal effect (3) by exactly the confounding contribution ($2 \cdot 1 = 2$, i.e., $2A$'s coefficient in X times 5's coefficient scaled appropriately; see the appendix for the general formula). Figure 24 confirms both values on simulated data: an observational regression recovers slope ≈ 5.03, while directly simulating the intervention (fixing X and averaging fresh A, U_Y draws) recovers slope ≈ 3.00.

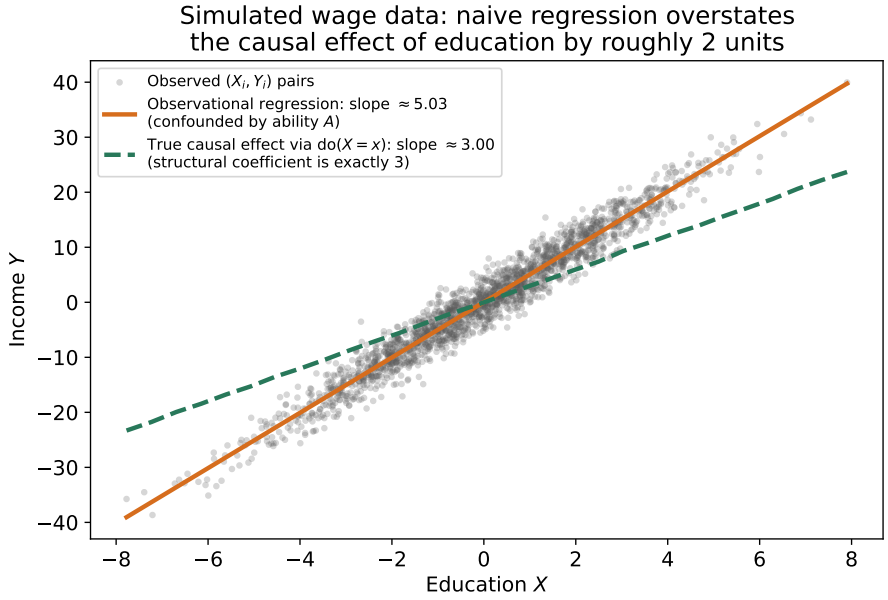


Figure 24: Simulated data from the wage SCM: the observational regression line is biased upward relative to the true causal (interventional) relationship.

Worked Problem 29.2 (A numeric counterfactual)**.** Suppose we observe one specific individual with $A = 1$ (a fixed, known ability level for this person) and record $U_X = 0.5$, $U_Y = -0.3$ exactly

(abduction step: fully determined since we treat A, U_X, U_Y as known for this illustration). Their actual education was $X = 2(1) + 0.5 = 2.5$ and actual income was $Y = 3(2.5) + 5(1) - 0.3 = 12.2$. *Counterfactual question*: what would this same person's income have been with 2 additional years of education, i.e., $X' = 4.5$, *holding their personal ability and personal luck (U_Y) fixed*? Action: set $X := 4.5$. Prediction: $Y' = 3(4.5) + 5(1) + (-0.3) = 18.2$. The counterfactual effect for this individual is $18.2 - 12.2 = 6.0$, exactly 3×2 additional years — because the structural coefficient on X is homogeneous (3) across individuals in this simple linear model; in a model with individual-level effect heterogeneity (coefficients depending on A or U), different individuals could have different counterfactual gains, a theme central to Session 10's treatment-effect heterogeneity.

Worked Problem 29.3 (Revisiting kidney stones as an SCM)**.** Model Session 7's kidney-stone study with Z (stone severity), X (treatment choice, A or B), Y (recovery), and structural equations $X := g(Z, U_X)$ (physicians' choice depends on severity) and $Y := h(X, Z, U_Y)$ (recovery depends on both treatment and severity). The naive observational comparison mixes Z's direct effect on Y with X's effect via the same fork logic as the wage example; Theorem 27.2 says the correct interventional quantity $P(y \mid \mathrm{do}(x))$ requires the factor $P(x \mid z, u_x)$ to be dropped and Z's marginal distribution to be kept as-is — which, when we compute it explicitly in Session 9, will turn out to be exactly the stratify-by-severity calculation the BMJ statisticians already performed by hand in 1994, now justified by a formal theorem rather than by good statistical instinct alone.

Additional Worked Examples (Instructor Supplement)

Worked Problem 29.4 (Truncated factorization on a discrete (non-Gaussian) SCM)**.** Every numeric example above used a linear-Gaussian SCM. Theorem 27.2 applies equally to discrete structural equations implemented via CPTs. Let A (market boom) have $P(A{=}1) = 0.3$; let advertising campaign X have $P(X{=}1 \mid A{=}1) = 0.9$, $P(X{=}1 \mid A{=}0) = 0.2$; and let high sales Y have $P(Y{=}1 \mid X{=}1, A{=}1) = 0.9$, $P(Y{=}1 \mid X{=}1, A{=}0) = 0.6$, $P(Y{=}1 \mid X{=}0, A{=}1) = 0.7$, $P(Y{=}1 \mid X{=}0, A{=}0) = 0.2$. Compute the true causal effect of the campaign and compare it to the naive observational contrast.

Solution (interventional, via Theorem 27.2). Drop the factor $P(x \mid a)$ and fix X:

$$P(Y{=}1 \mid \mathrm{do}(X{=}1)) = \sum_a P(Y{=}1 \mid X{=}1, a)P(a) = 0.9(0.3) + 0.6(0.7) = 0.27 + 0.42 = \mathbf{0.69},$$

$$P(Y{=}1 \mid \mathrm{do}(X{=}0)) = 0.7(0.3) + 0.2(0.7) = 0.21 + 0.14 = \mathbf{0.35}.$$

True causal effect $= 0.69 - 0.35 = \mathbf{0.34}$.

Solution (observational). First $P(X{=}1) = 0.9(0.3) + 0.2(0.7) = 0.27 + 0.14 = 0.41$. By Bayes' rule, $P(A{=}1 \mid X{=}1) = 0.9(0.3)/0.41 \approx 0.6585$, and $P(A{=}1 \mid X{=}0) = 0.1(0.3)/0.59 \approx 0.0508$. Then

$$P(Y{=}1 \mid X{=}1) \approx 0.9(0.6585) + 0.6(0.3415) \approx 0.5927 + 0.2049 = \mathbf{0.7976},$$

$$P(Y{=}1 \mid X{=}0) \approx 0.7(0.0508) + 0.2(0.9492) \approx 0.0356 + 0.1898 = \mathbf{0.2254}.$$

Naive difference $\approx 0.7976 - 0.2254 = \mathbf{0.5722}$, badly overstating the true effect (0.34) — the same confounding-bias phenomenon as Example 26.3, now for a fully categorical SCM where every quantity is an ordinary CPT rather than a Gaussian regression coefficient.

Worked Problem 29.5 (Abduction is not always a single point: a non-invertible discrete mechanism)**.** The three-step counterfactual procedure's abduction step was exact and deterministic in every worked example above because each structural equation was *invertible* in its noise given the

parents. This need not hold. Let $X := \mathbb{1}[U_1{=}1 \text{ or } U_2{=}1]$, with U_1, U_2 i.i.d. Bernoulli(0.5), independent of everything else. Suppose we observe $X{=}1$ but never see U_1, U_2 individually. What does abduction give us?

Solution. The prior over (U_1, U_2) is uniform over 4 equally likely combinations, each probability 0.25. Three of the four combinations are consistent with $X{=}1$: $(1,0), (0,1), (1,1)$ (only $(0,0)$ gives $X{=}0$). Since all three had equal prior probability, each has posterior probability $0.25/0.75 = \mathbf{1/3}$ given $X{=}1$:

$$P(U_1{=}1, U_2{=}0 \mid X{=}1) = P(U_1{=}0, U_2{=}1 \mid X{=}1) = P(U_1{=}1, U_2{=}1 \mid X{=}1) = \tfrac{1}{3}.$$

Unlike every linear-Gaussian example above, abduction here does *not* pin down a single value of the exogenous noise — it yields a genuine three-point *distribution* over (U_1, U_2). A downstream counterfactual computation (e.g., "what would some effect of X have been under a different intervention, for this same unit") must then *average* over this posterior, rather than plug in one exact noise value as the prediction step did in Exercise 8.2. Non-invertible structural equations — common whenever a discrete variable is a many-to-one function of its causes — are the norm rather than the exception outside idealized linear models, and this is precisely why real implementations of counterfactual reasoning (e.g., in probabilistic programming systems) must track a full posterior over $\mathbf{U}$, not a point estimate.

Appendix: General Confounding-Bias Formula for Linear SCMs

For the general linear SCM $X = aA + U_X$, $Y = bX + cA + U_Y$ (all noises independent, unit variance, A unit variance), the observational regression slope of Y on X is $\dfrac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(X)} = \dfrac{a(ab+c)+b}{a^2+1} = b + \dfrac{ac}{a^2+1}$, while the true causal effect is exactly b (from $\mathrm{do}(X = x)$: $\mathbb{E}[Y \mid \mathrm{do}(x)] = bx + c\mathbb{E}[A] = bx$ when $\mathbb{E}[A] = 0$). The bias term $\dfrac{ac}{a^2+1}$ vanishes exactly when $a = 0$ (no confounder $\to X$ edge) or $c = 0$ (no confounder $\to Y$ edge) — confirming that confounding bias requires *both* arrows of the fork to be present, consistent with the graphical picture throughout this session. Plugging in $a = 2, b = 3, c = 5$ recovers the worked example's bias of $\frac{10}{5} = 2$, i.e., observational slope $3 + 2 = 5$, matching both the hand calculation and the simulation.

Connections and Takeaways

We now have a mathematical object — the SCM — in which "the effect of an intervention" and even "what would have happened to this specific unit" are well-defined quantities, computed by the truncated factorization formula (Theorem 27.2) directly from the same graphical and computational machinery built in Sessions 2–3. This resolves Session 7's conceptual gap in principle.

But notice the catch in every worked example above: we always assumed we had the *full structural equations*, including the confounder A and its exact numerical role. In practice we almost never observe every relevant confounder, let alone know its functional form. **Session 9** asks the sharper, practitioner-relevant question: if all we are given is an observational dataset plus a causal *graph* (no equations, and possibly some variables, like A, unmeasured) can we still compute $P(y \mid \mathrm{do}(x))$ using only the pieces we can actually observe? The answer

will be: sometimes yes, via a checkable graphical criterion — and sometimes provably no.

Session 9: Intervention and Identification: Do-Calculus

Opening Motivation

Session 8's truncated factorization formula (Theorem 8.3) computes $P(y \mid \text{do}(x))$ — *but only if we already have the full set of structural equations*, confounders and all. In practice we almost never observe every confounder: no dataset records "ability" directly, and no observational cancer study directly measures every genetic predisposition. The practitioner's real question is sharper: given only an observational dataset and a *graph* (which may include boxes for variables we know exist causally but cannot measure), can we still write $P(y \mid \text{do}(x))$ as a formula using only observable quantities? This is the *identification* problem, and today's answer — the backdoor criterion, the front-door criterion, and the complete rules of do-calculus — turns Session 7's confounding heuristics into checkable theorems.

30 The Identification Problem

Definition 30.1 (Identifiability)**.** A causal effect $P(y \mid \text{do}(x))$ is *identifiable* from a causal graph G if it can be expressed as a function of the observational distribution $P(\mathbf{v})$ over the observed variables alone — computable without knowledge of the specific structural equations or unobserved-variable distributions, using only G's structure and observational data.

31 The Backdoor Criterion

Definition 31.1 (Backdoor path)**.** A path from X to Y is a *backdoor path* if it starts with an arrow *into* X (i.e., it is not a directed path out of X).

Definition 31.2 (Backdoor criterion)**.** A set of variables Z satisfies the backdoor criterion relative to (X, Y) if: (i) no node in Z is a descendant of X, and (ii) Z blocks every backdoor path between X and Y (in the sense of d-separation, Session 2).

Theorem 31.3 (Backdoor adjustment)**.** *If Z satisfies the backdoor criterion relative to (X, Y), then $P(y \mid \text{do}(x))$ is identifiable and equals*

$$P\big(y \mid \text{do}(x)\big) = \sum_{z} P(y \mid x, z)\, P(z).$$

This is exactly the stratify-and-reweight calculation Session 7's real-world investigators performed by hand on the Berkeley and kidney-stone data — now justified as the unique correct formula whenever Z satisfies a checkable graphical condition, rather than a statistical instinct to "control for confounders."

Worked Problem 31.4 (Formally resolving the kidney-stone paradox)**.** In the SCM $X := g(Z, U_X)$, $Y := h(X, Z, U_Y)$ from Session 8, stone severity Z satisfies the backdoor criterion for (X, Y): it is not a descendant of treatment X (severity is determined before treatment is chosen), and it blocks the only backdoor path $X \leftarrow Z \rightarrow Y$. Applying Theorem 31.3 to the *real* data from Session 7 (Figure 25): $P(Z{=}\text{small}) = 357/700 = 0.510$, $P(Z{=}\text{large}) = 343/700 = 0.490$, so

$$P(\text{recover} \mid \text{do}(A)) = 0.931(0.510) + 0.730(0.490) = 0.8325, \qquad P(\text{recover} \mid \text{do}(B)) = 0.867(0.510) + 0.688(0.490) =$$

Treatment A's adjusted causal effect (83.3%) now correctly exceeds Treatment B's (77.9%), matching what both size-stratified subgroups already showed individually in Session 7 — Theorem 31.3 is precisely the formula that makes this the mathematically *correct* way to combine the two strata back together, rather than the naive (unweighted-by-mechanism) aggregate that produced the original paradox.

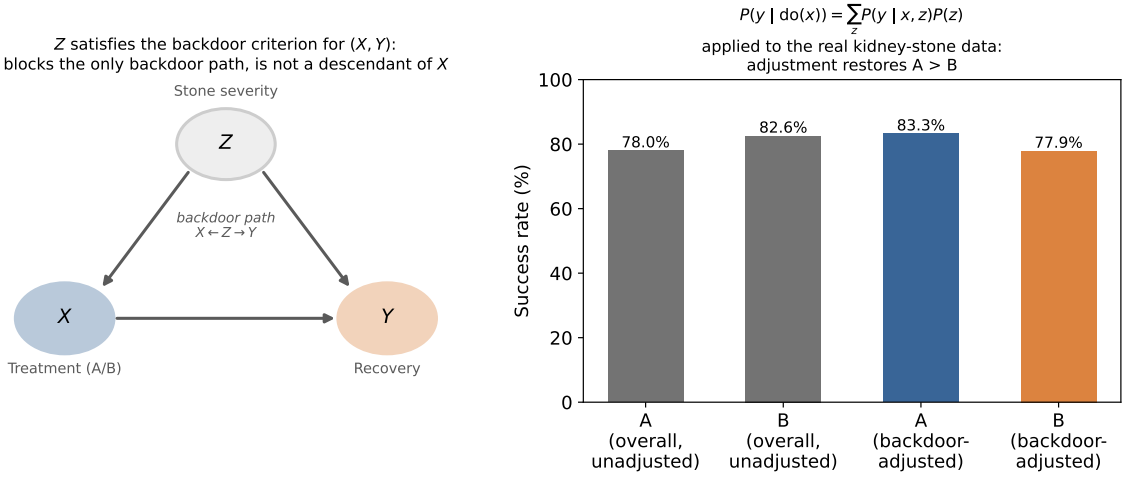


Figure 25: Left: stone severity Z satisfies the backdoor criterion. Right: the backdoor adjustment formula applied to the real 1986 data restores the correct ranking of treatments.

32 The Front-Door Criterion: When Adjustment Is Impossible but Identification Still Succeeds

The backdoor criterion requires *measuring* enough of the confounding structure to block every backdoor path. What if the confounder itself is unmeasurable?

Example 32.1 (Smoking, tar, and cancer)**.** Consider $X =$ smoking, $Y =$ lung cancer, with a hypothesized unmeasured genetic factor U that predisposes people both to smoke and (independently) to develop cancer — a classic historical objection to the smoking-cancer causal claim. Suppose additionally that smoking's effect on cancer is entirely mediated by tar deposits in the lungs, M: $X \to M \to Y$, with U affecting X and Y directly but *not* M (Figure 26).

Definition 32.2 (Front-door criterion)**.** A set of variables M satisfies the front-door criterion relative to (X, Y) if: (i) M intercepts all directed paths from X to Y; (ii) there is no unblocked backdoor path from X to M; (iii) every backdoor path from M to Y is blocked by X.

Theorem 32.3 (Front-door adjustment)**.** *If M satisfies the front-door criterion relative to (X, Y), then $P(y \mid \mathrm{do}(x))$ is identifiable and equals*

$$P(y \mid \mathrm{do}(x)) = \sum_{m} P(m \mid x) \sum_{x'} P(y \mid x', m)\, P(x').$$

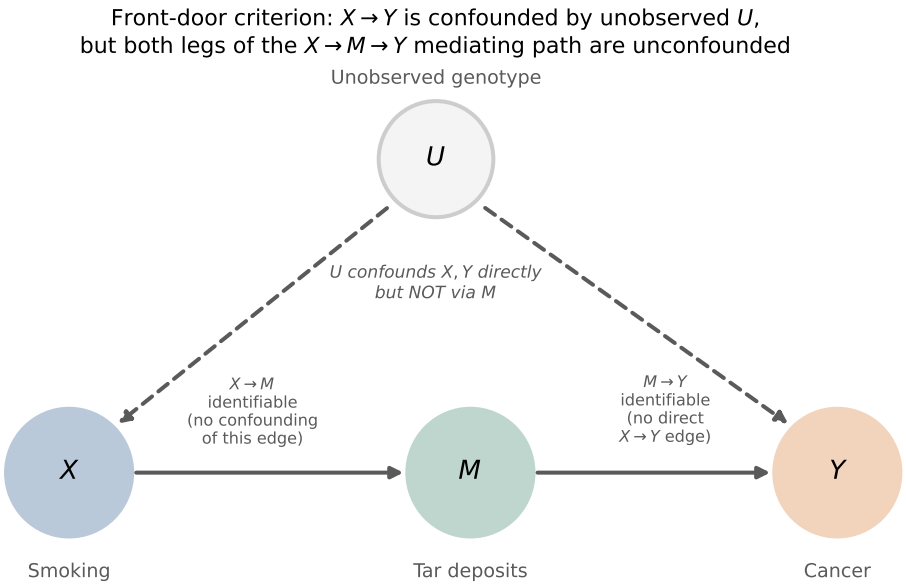


Figure 26: The front-door configuration: $X \to Y$ is confounded by unobserved U, but the mediating path through M is unconfounded in both legs.

Intuition

The formula estimates the effect in two unconfounded legs and chains them: $P(m \mid x)$ is the (unconfounded, by condition ii) effect of smoking on tar; the inner sum $\sum_{x'} P(y \mid x', m)P(x')$ estimates the (unconfounded once we further adjust for X, by condition iii) effect of tar on cancer, holding the confounding influence of U on Y fixed by averaging over X's marginal distribution rather than conditioning on the specific x we started with. The genuinely surprising part is that the unmeasured, un-adjustable confounder U between X and Y never needs to be measured at all — it is defeated by routing around it through a mediator both of whose legs happen to be individually unconfounded.

33 Do-Calculus: The Complete Rules

Backdoor and front-door are the two most commonly invoked special cases of a fully general calculus for manipulating do(·) expressions, built entirely from d-separation applied to mutilated graphs. Write $G_{\overline{X}}$ for G with X's incoming edges removed (Session 8's mutilation) and $G_{\underline{X}}$ for G with X's outgoing edges removed.

Theorem 33.1 (Rules of do-calculus, Pearl 1995)**.** *For disjoint variable sets X, Y, Z, W:*

Rule 1 (insertion/deletion of observations): $P(y \mid \mathrm{do}(x), z, w) = P(y \mid \mathrm{do}(x), w)$ *if* $(Y \perp\!\!\!\perp_d Z \mid X, W)$ *in* $G_{\overline{X}}$.

Rule 2 (action/observation exchange): $P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), z, w)$ *if* $(Y \perp\!\!\!\perp_d Z \mid X, W)$ *in* $G_{\overline{X}\underline{Z}}$.

Rule 3 (insertion/deletion of actions): $P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), w)$ *if* $(Y \perp\!\!\!\perp_d Z \mid X, W)$ *in* $G_{\overline{X}\,\overline{Z(W)}}$, *where* $Z(W)$ *excludes any* Z*-nodes that are ancestors of* W *in* $G_{\overline{X}}$.

These three rules are sound and complete*: a causal effect is identifiable if and only if it can be reduced to a purely observational expression by some finite sequence of applications of Rules 1–3 (together with ordinary probability axioms).*

We state completeness without proof (it is a substantial graph-theoretic result; Shpitser & Pearl, 2006, give an algorithm that always finds the reduction when one exists). What matters for practice is the intuition behind each rule: every one of them is exactly Session 2's d-separation criterion, applied not to the original graph but to a graph mutilated by the actions already performed — do-calculus adds nothing conceptually new to d-separation, it only asks us to check separation in the right (possibly surgically modified) graph for each manipulation we want to justify. Both Theorem 31.3 and Theorem 32.3 can be, and originally were, derived as short sequences of these three rules; we do this explicitly for the backdoor case in the appendix.

34 Non-Identifiability: When No Formula Exists

Example 34.1 (The bow-arc graph)**.** Let U be an unobserved common cause of both X and Y, with also a direct edge $X \to Y$, and no measured mediator at all (Figure 27). No set of *observed* variables can satisfy the backdoor criterion (the only backdoor path $X \leftarrow U \to Y$ passes through the unmeasured U, and there are no other measured variables to substitute), and the front-door criterion requires a mediator M intercepting the $X \to Y$ edge, which does not exist here either.

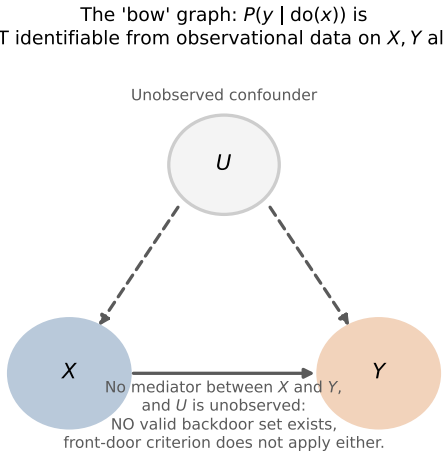


Figure 27: A structure for which $P(y \mid \mathrm{do}(x))$ is provably not identifiable from $P(x, y)$ alone.

Theorem 34.2 (Non-identifiability of the bow-arc graph)**.** *For the graph in Figure 27, no application of do-calculus (Theorem 33.1) can reduce* $P(y \mid \mathrm{do}(x))$ *to an expression in* $P(x, y)$ *alone; there exist two distinct SCMs sharing this graph and inducing the identical observational distribution* $P(x, y)$ *but different interventional distributions* $P(y \mid \mathrm{do}(x))$.

This is not a failure of cleverness — it is a mathematical impossibility, and recognizing it (rather than reporting a confidently wrong number) is itself the valuable skill this session teaches. When a

causal graph implies non-identifiability, the only paths forward are: collect more data (measure a proxy for U, breaking the bow), run a randomized experiment (Session 10's benchmark case, which sidesteps identification entirely by design), or bound the effect rather than pin it down exactly (partial identification, beyond our scope here).

35 Worked Examples

Worked Problem 35.1 (Checking the backdoor criterion carefully)**.** In the kidney-stone graph (Figure 25), verify condition (i) of the backdoor criterion: is stone severity Z a descendant of treatment X? *Solution:* No — severity is a biological fact determined before any treatment decision, so $Z \notin \mathrm{de}(X)$, and condition (i) holds. This matters because adjusting for a *post-treatment* variable (e.g., a side-effect of treatment that also affects recovery) can introduce exactly the collider bias warned about in Sessions 2 and 7 — the backdoor criterion's descendant restriction is precisely what rules this mistake out.

Worked Problem 35.2 (An online-experimentation graph, resolving Session 7's ad vignette)**.** Recall Session 7's marketing example: ad exposure X, conversion Y, confounded by purchase intent Z (which drives both the targeting algorithm's choice to show the ad and the baseline conversion probability). Z satisfies the backdoor criterion exactly as stone severity did. Using the hypothetical numbers from Session 7 (25% conversion for high-intent customers regardless of ad, 5% for low-intent, 70% of ad-shown customers high-intent vs. 20% of not-shown), and treating overall intent prevalence as $P(Z{=}\text{high}) = 0.4$ across the full customer base: $P(Y{=}1 \mid \mathrm{do}(X{=}1)) = 0.25(0.4) + 0.05(0.6) = 0.13$, and identically $P(Y{=}1 \mid \mathrm{do}(X{=}0)) = 0.25(0.4) + 0.05(0.6) = 0.13$ — the backdoor-adjusted effect of the ad is exactly zero, correctly recovering the fact (stated by construction in Session 7) that the ad had no true causal effect at all, in sharp contrast to the naive 19% vs. 9% comparison that ignored intent entirely.

Worked Problem 35.3 (Front-door numerically)**.** Suppose (illustrative numbers) $P(M{=}1 \mid X{=}1) = 0.8$, $P(M{=}1 \mid X{=}0) = 0.2$ (smoking strongly increases tar), $P(X{=}1) = 0.4$ in the population, and $P(Y{=}1 \mid X{=}1, M{=}1) = 0.5$, $P(Y{=}1 \mid X{=}0, M{=}1) = 0.5$, $P(Y{=}1 \mid X{=}1, M{=}0) = 0.1$, $P(Y{=}1 \mid X{=}0, M{=}0) = 0.1$ (cancer risk depends only on tar, not directly on smoking or on X once M is fixed — reflecting no direct $X \to Y$ edge and no residual U leakage into this conditional). Then $\sum_{x'} P(Y{=}1 \mid x', M{=}1)P(x') = 0.5(0.4) + 0.5(0.6) = 0.5$ and similarly $\sum_{x'} P(Y{=}1 \mid x', M{=}0)P(x') = 0.1$. So $P(Y{=}1 \mid \mathrm{do}(X{=}1)) = P(M{=}1 \mid X{=}1)(0.5) + P(M{=}0 \mid X{=}1)(0.1) = 0.8(0.5) + 0.2(0.1) = 0.42$, versus $P(Y{=}1 \mid \mathrm{do}(X{=}0)) = 0.2(0.5) + 0.8(0.1) = 0.18$ — a genuine causal effect of smoking on cancer of 0.24, recovered entirely without ever measuring the confounding genotype U.

Additional Worked Examples (Instructor Supplement)

Worked Problem 35.4 (Backdoor adjustment with a genuinely two-dimensional adjustment set)**.** Every backdoor example above used a single confounder Z. Let $X \to Y$ be confounded by *two* independent variables: $Z_1 \to X$, $Z_1 \to Y$, $Z_2 \to X$, $Z_2 \to Y$, with $P(Z_1{=}1) = 0.4$, $P(Z_2{=}1) = 0.6$ (independent), and $P(Y{=}1 \mid X{=}1, z_1, z_2) = 0.5 + 0.1z_1 + 0.1z_2$, $P(Y{=}1 \mid X{=}0, z_1, z_2) = 0.3 + 0.1z_1 + 0.1z_2$ (so the true effect of X is a constant 0.2 regardless of z_1, z_2). Confirm that $\{Z_1, Z_2\}$ recovers this exactly.

Solution. By independence, $P(z_1, z_2)$ for the four combinations is $P(0,0){=}0.6(0.4){=}0.24$, $P(1,0){=}0.4(0.4){=}0.16$ $P(0,1){=}0.6(0.6){=}0.36$, $P(1,1){=}0.4(0.6){=}0.24$ (summing to 1). Then

$$P(Y{=}1 \mid \mathrm{do}(X{=}1)) = 0.5(0.24){+}0.6(0.16){+}0.6(0.36){+}0.7(0.24) = 0.12{+}0.096{+}0.216{+}0.168 = \mathbf{0.60},$$

$P(Y{=}1 \mid \mathrm{do}(X{=}0)) = 0.3(0.24)+0.4(0.16)+0.4(0.36)+0.5(0.24) = 0.072+0.064+0.144+0.12 = \mathbf{0.40}$.

The recovered effect $0.60 - 0.40 = \mathbf{0.20}$ matches the true, constant structural effect exactly. The backdoor adjustment formula $P(y \mid \mathrm{do}(x)) = \sum_{z_1,z_2} P(y \mid x, z_1, z_2)P(z_1, z_2)$ generalizes immediately to a vector-valued Z; the only practical change is that the sum now ranges over every joint configuration of all adjustment variables, which is exactly why Session 10 will need machine learning (rather than a hand-built table) once Z has dozens of real-valued covariates instead of two binary ones.

Worked Problem 35.5 (A valid backdoor set need not be minimal)**.** Consider $Z_2 \to Z_1$, $Z_1 \to X$, $Z_1 \to Y$, $X \to Y$ (Z_2 is an ancestor of Z_1 but has no other connection to X or Y). Does $\{Z_1\}$ satisfy the backdoor criterion? Does $\{Z_1, Z_2\}$?

Solution. $\{Z_1\}$: not a descendant of X, and it blocks the only backdoor path $X \leftarrow Z_1 \to Y$ (a fork), so $\{Z_1\}$ is valid — and minimal. $\{Z_1, Z_2\}$: Z_2 is also not a descendant of X (it is further upstream), and the backdoor path is still blocked by $Z_1 \in \{Z_1, Z_2\}$ regardless of whether Z_2 is additionally included, so $\{Z_1, Z_2\}$ is *also* a valid backdoor set — just not a minimal one. Including a harmless ancestor of an already-sufficient adjustment variable never breaks validity; it can only cost estimation efficiency later (Session 10: more covariates to adjust for means more parameters to estimate from finite data), never correctness. This is the opposite lesson from Exercise 9.1's $\{Z, W\}$ (which included a genuine *descendant* of X and thereby broke condition (i)) — the backdoor criterion is sensitive to which *kind* of extra variable you add, not merely to how many you add.

Worked Problem 35.6 (An explicit application of Rule 1)**.** Let $X \to Y$ and also $X \to W$, with no edge between W and Y at all (W is an unrelated side-effect of X). Show $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x))$ using Rule 1.

Solution. Rule 1 requires $Y \perp\!\!\!\perp_d W \mid X$ in $G_{\overline{X}}$ (delete edges into X; here X has none, so $G_{\overline{X}} = G$). In G, the only path between W and Y is $W \leftarrow X \to Y$, a fork with fork-node X; since X is in the conditioning set, this path is blocked, and there is no other path (by assumption W, Y share no direct edge), so $Y \perp\!\!\!\perp_d W \mid X$ holds. Rule 1 therefore licenses dropping w from the conditioning set: $P(y \mid \mathrm{do}(x), w) = P(y \mid \mathrm{do}(x))$. Practically, this says an irrelevant post-treatment side effect of X — one that does not itself influence Y — can always be safely ignored when computing Y's response to the intervention, without needing to measure or adjust for it at all.

Appendix: Deriving Backdoor Adjustment from Do-Calculus Rules

We derive Theorem 31.3 from Theorem 33.1 for Z satisfying the backdoor criterion. Start from $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid \mathrm{do}(x), z)P(z \mid \mathrm{do}(x))$ (law of total probability). Since Z contains no descendants of X, intervening on X cannot affect Z's distribution, so $P(z \mid \mathrm{do}(x)) = P(z)$ (a direct application of Rule 3, since $X \perp\!\!\!\perp_d Z$ in $G_{\overline{X}}$ trivially — Z has no path from X once X's outgoing edges to any descendant of Z are irrelevant and Z is not a descendant of X by assumption). For the remaining term, since Z blocks every backdoor path, $Y \perp\!\!\!\perp_d X \mid Z$ holds in $G_{\underline{X}}$ (removing X's outgoing edges leaves only backdoor paths, all blocked by Z), so by Rule 2, $P(y \mid \mathrm{do}(x), z) = P(y \mid x, z)$. Substituting both simplifications gives $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z)$, exactly Theorem 31.3.

Connections and Takeaways

We can now determine, directly from a graph, *whether* $P(y \mid \mathrm{do}(x))$ is a well-defined function of observational data (backdoor, front-door, or in general via do-calculus) and, when it is, write that function down explicitly. This is a complete answer to the identification question posed at the top of this session.

What we have *not* done is estimate any of these quantities from finite, noisy, real data — every formula above was applied either to population-level quantities or to a dataset treated as if it perfectly revealed $P(y \mid x, z)$ and $P(z)$. **Session 10**, the course's capstone, closes this last gap: given real observational data, how do we actually build good estimators of the identified causal effect — propensity scores, inverse propensity weighting, doubly robust estimation, and machine-learning-based treatment-effect estimators — and how does the special case of a randomized experiment (where identification is guaranteed by design) connect back to everything built since Session 7?

Session 10: Causal Effects and Machine Learning

Opening Motivation

Session 9 told us exactly which quantity to compute — $P(y \mid \mathrm{do}(x)) = \sum_z P(y \mid x, z)P(z)$ under the backdoor criterion — but every worked example assumed the relevant conditional distributions were simply known. Real data is finite, noisy, and high-dimensional: Z might be dozens of covariates, not one categorical variable, and $P(y \mid x, z)$ is exactly the kind of function machine learning is built to estimate. This capstone session connects the identification theory of Sessions 7–9 to the practitioner's actual workflow: build estimators, using tools students already know (regression, classification, the reweighting ideas from Session 6's importance sampling), that target precisely the causal quantity Session 9 proved was well-defined.

36 From Identification to Estimation

Suppose X is binary treatment, Y the outcome, and Z a (possibly high-dimensional) set of covariates satisfying the backdoor criterion. The quantity of interest is usually the **average treatment effect**,

$$\mathrm{ATE} = \mathbb{E}\big[Y \mid \mathrm{do}(X{=}1)\big] - \mathbb{E}\big[Y \mid \mathrm{do}(X{=}0)\big] = \mathbb{E}_Z\Big[\mathbb{E}[Y \mid X{=}1, Z] - \mathbb{E}[Y \mid X{=}0, Z]\Big],$$

the second equality following directly from Theorem 9.1's backdoor adjustment. Three broad estimation strategies exist, each exploiting a different piece of this formula.

36.1 Outcome Regression

Directly estimate $\hat{\mu}_1(z) = \widehat{\mathbb{E}}[Y \mid X{=}1, Z{=}z]$ and $\hat{\mu}_0(z) = \widehat{\mathbb{E}}[Y \mid X{=}0, Z{=}z]$ (any regression model — linear regression, gradient boosting, a neural network) using the observed data, and average their difference:

$$\widehat{\mathrm{ATE}}_{\mathrm{reg}} = \frac{1}{n}\sum_{i=1}^{n}\big[\hat{\mu}_1(z_i) - \hat{\mu}_0(z_i)\big].$$

This is the most direct payoff of Session 5's learning machinery: it is ordinary supervised learning, applied twice (once per treatment arm), then combined via Session 9's adjustment formula. Its weakness: if $\hat{\mu}_1, \hat{\mu}_0$ are misspecified (wrong functional form), the ATE estimate is biased, with no diagnostic built in.

36.2 Propensity Scores and Inverse Propensity Weighting

Definition 36.1 (Propensity score)**.** $e(z) = P(X{=}1 \mid Z{=}z)$, the probability of receiving treatment given covariates.

Theorem 36.2 (Balancing property)**.** *$X \perp\!\!\!\perp Z \mid e(Z)$, and consequently $Y(0), Y(1) \perp\!\!\!\perp X \mid e(Z)$ whenever $Y(0), Y(1) \perp\!\!\!\perp X \mid Z$ (the backdoor / "no unmeasured confounding" condition, in potential-outcomes notation). Adjusting for the single scalar $e(Z)$ suffices — adjusting for all of Z is sufficient but not necessary.*

This is exactly the phenomenon Session 6's importance sampling relied on: we cannot easily sample from the "everyone treated" world directly, so we reweight the treated units we *did* observe to stand in for that counterfactual population.

Definition 36.3 (Inverse propensity weighting)**.**

$$\widehat{\mathrm{ATE}}_{\mathrm{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{T_i Y_i}{\hat{e}(z_i)} - \frac{(1-T_i)Y_i}{1-\hat{e}(z_i)}\right].$$

Each treated unit is reweighted by $1/\hat{e}(z_i)$ — units who were *unlikely* to be treated given their covariates but were treated anyway are up-weighted, since they are rare and informative "surprises" standing in for many similar untreated units; this is precisely Session 6's importance-sampling correction $P(\mathbf{x})/Q(\mathbf{x})$, with Q here being "what treatment assignment actually looked like" and P being "what it would look like if everyone were treated."

36.3 Doubly Robust Estimation

Theorem 36.4 (Double robustness)**.** *Define the augmented IPW (AIPW) estimator*

$$\widehat{\mathrm{ATE}}_{AIPW} = \frac{1}{n}\sum_{i}\left[\hat{\mu}_1(z_i) - \hat{\mu}_0(z_i) + \frac{T_i(Y_i - \hat{\mu}_1(z_i))}{\hat{e}(z_i)} - \frac{(1-T_i)(Y_i - \hat{\mu}_0(z_i))}{1-\hat{e}(z_i)}\right].$$

This estimator is consistent for the true ATE if either *the outcome models $\hat{\mu}_1, \hat{\mu}_0$ are correctly specified* or *the propensity model $\hat{e}$ is correctly specified — it needs only one of the two to be right, not both.*

Intuition

Think of AIPW as the outcome-regression estimate plus a bias-correction term. If the outcome model is exactly right, the correction terms have expectation zero and add only noise. If the outcome model is wrong but the propensity model is right, the correction term's reweighting exactly cancels the outcome model's bias. "Two chances to get it right" is the standard, and accurate, shorthand.

37 Heterogeneous Effects and Meta-Learners

The ATE averages over the whole population, but a marketing team choosing whom to target, or a clinician choosing whom to treat, needs the **conditional average treatment effect**, $\mathrm{CATE}(z) = \mathbb{E}[Y \mid \mathrm{do}(X{=}1), Z{=}z] - \mathbb{E}[Y \mid \mathrm{do}(X{=}0), Z{=}z]$. *Meta-learners* repurpose ordinary supervised learning models to estimate $\mathrm{CATE}(z)$:

- **T-learner**: fit two separate models $\hat{\mu}_1(z)$, $\hat{\mu}_0(z)$ (one per treatment arm, exactly as in outcome regression above) and set $\widehat{\mathrm{CATE}}(z) = \hat{\mu}_1(z) - \hat{\mu}_0(z)$.
- **S-learner**: fit a single model $\hat{\mu}(z, x)$ with treatment as an ordinary input feature, and set $\widehat{\mathrm{CATE}}(z) = \hat{\mu}(z, 1) - \hat{\mu}(z, 0)$.
- **X-learner**: impute individual-level "pseudo-effects" using the T-learner's cross-arm predictions, then regress these pseudo-effects on Z — generally more robust when treatment groups are very unbalanced in size.

38 Worked Examples

Worked Problem 38.1 (A confounded targeting simulation, all estimators checked against ground truth)**.** Simulate $n = 4000$ customers with engagement score $X \sim \text{Uniform}(0, 1)$ (a confounder, renamed Z in the general notation above), true propensity $e(X) = \text{sigmoid}(4X - 2)$ (highly-engaged customers are much more likely to receive a targeted offer T), and true heterogeneous treatment effect $\text{CATE}(x) = 2 + 3x$ (the offer helps more-engaged customers more), so the true $\text{ATE} = \mathbb{E}[2+3X] = 3.5$. Figure 28 confirms adequate propensity-score overlap between treated and control groups (a necessary condition for any of these estimators to be reliable — if some region of Z contained *only* treated units, no untreated comparison would exist there at all). Figure 29 compares five estimates: a naive difference in means (4.47, badly biased upward because engaged customers both received the offer more often *and* had higher baseline outcomes even without it — exactly Session 7's confounding pattern), outcome regression (3.52), IPW (3.52), AIPW (3.52), and a benchmark from a hypothetical randomized experiment on the same population (3.53). All four adjustment-based estimators land within statistical noise of the true value 3.51; only the naive comparison is off, and it is off by an amount (≈ 1.0, about 28% relative bias) that would matter enormously for a real business decision.

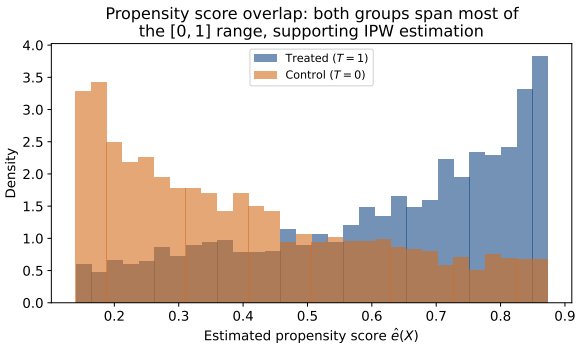


Figure 28: Propensity score overlap between treated and control groups.

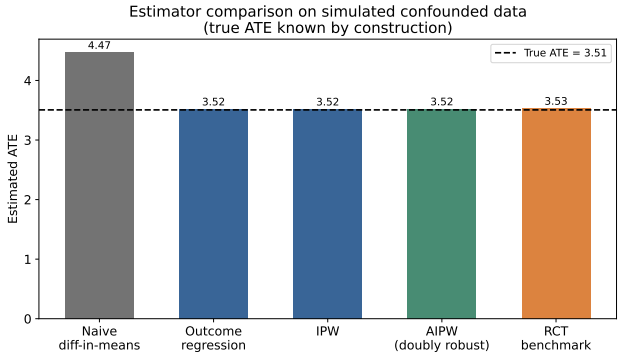


Figure 29: All confounding-adjusted estimators recover the true ATE; the naive comparison does not.

Worked Problem 38.2 (Recovering who benefits most: the T-learner)**.** Using the same simulated data, fit two separate linear regressions of Y on engagement X — one using only treated customers, one using only untreated customers — and take their difference as $\widehat{\text{CATE}}(x)$. Figure 30 shows the result tracks the true $\text{CATE}(x) = 2 + 3x$ closely across the full range of engagement scores. This is directly actionable: a firm with a limited marketing budget should target the *highest*-engagement customers first, not because they are easiest to reach, but because the estimated causal effect of the offer is largest for them — a decision that a single average treatment effect number could never have supported.

Worked Problem 38.3 (Why the RCT benchmark needs no identification argument at all)**.** In the simulated randomized experiment used as a benchmark above, treatment was assigned by a fair coin flip *independent of* X: $e(x) = 0.5$ for everyone. Because assignment does not depend on X (or anything else), $Z = \emptyset$ trivially satisfies the backdoor criterion — there is no confounding to adjust for, so the naive difference in means *is* already the correct estimator of the ATE, with

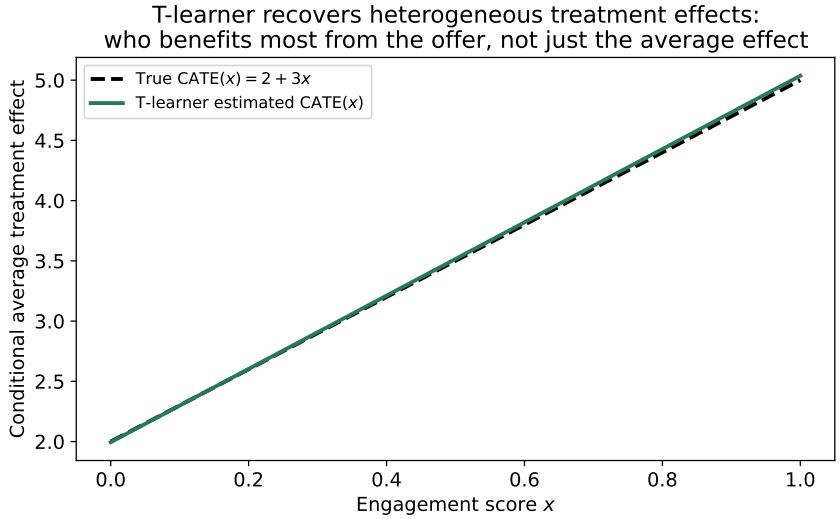


Figure 30: T-learner estimated CATE closely tracks the true (simulated) heterogeneous treatment effect.

no need for the propensity or outcome models built above at all. This is the deepest connection this course draws between Sessions 7–10 and the classical A/B-testing framework familiar from any introductory statistics course: **a randomized experiment is not a different kind of causal inference — it is the special case where the identification problem Session 9 solved with real analytical work is instead solved automatically by the experimental design itself**, since randomization guarantees $e(Z)$ is known and constant by construction, making Z trivially sufficient for backdoor adjustment.

Additional Worked Examples (Instructor Supplement)

Worked Problem 38.4 (Quantifying IPW's variance blow-up from a single extreme-propensity unit)**.** Session 9's overlap discussion is qualitative; here is a concrete instance. A treated unit i with $Y_i = 10$ has propensity $\hat{e}(z_i) = 0.05$ (rare: this unit's covariates made treatment unlikely). Its IPW contribution is $Y_i/\hat{e}(z_i) = 10/0.05 = \mathbf{200}$. Compare to a "typical" treated unit with the same outcome $Y = 10$ but $\hat{e}(z) = 0.5$: its contribution is $10/0.5 = \mathbf{20}$. The rare-propensity unit alone contributes $10\times$ as much to the running sum $\frac{1}{n}\sum_i[\cdots]$ as a typical unit does — meaning a single observation can swing the entire average as much as ten ordinary ones combined. This is precisely the mechanism behind the variance inflation warned about qualitatively above: near-violations of overlap do not merely add a little noise, they let individual data points dominate the estimator, which is why practitioners routinely trim or cap extreme propensity weights in applied work.

Worked Problem 38.5 (Why an S-learner without interaction terms can miss heterogeneity entirely)**.** Two covariate strata, low and high engagement, each half the population, have *true* effects $\text{CATE}(\text{low}) = \mu_1(\text{low}) - \mu_0(\text{low}) = 5 - 5 = 0$ and $\text{CATE}(\text{high}) = 5 - 2 = 3$ (so $\text{ATE} = 0.5(0) + 0.5(3) = 1.5$). Suppose an S-learner is fit with a purely *additive* functional form, $\hat{\mu}(z, x) = a + bz + cx$ — no $z \times x$ interaction term. Explain why this model cannot recover the true heterogeneous effects, and what it recovers instead.

Solution. With no interaction term, the model's estimated treatment effect $\hat{\mu}(z, 1) - \hat{\mu}(z, 0) = c$ is forced to be the *same constant for every* z — it structurally cannot output 0 for low-engagement customers and 3 for high-engagement customers simultaneously. Fit by least squares against data

generated from the true heterogeneous process, such a model will instead converge toward something close to the population-averaged effect, $c \approx \text{ATE} = 1.5$, applied uniformly to *both* groups — correctly capturing the overall average impact while completely erasing the fact that the true effect is 0 for one group and 3 for the other. This is the standard critique of S-learners: any regularization or model-family choice that shrinks or omits the treatment-covariate interaction will bias $\widehat{\text{CATE}}(z)$ toward the pooled ATE, understating genuine heterogeneity — precisely the failure mode a T-learner (Exercise 10.2) avoids by construction, since it fits two entirely separate models with no shared interaction term to shrink.

Worked Problem 38.6 (Why cross-fitting matters: an overfitting thought experiment)**.** Consider fitting an outcome model $\hat{\mu}_1$ on $n = 4$ treated units using a model flexible enough to have 4 free parameters (one per data point) — e.g., a lookup table. It achieves *zero* training residual by construction: $Y_i - \hat{\mu}_1(z_i) = 0$ for all 4 points. What happens to the AIPW correction term computed on this same data?

Solution. The correction term $\frac{T_i(Y_i - \hat{\mu}_1(z_i))}{\hat{e}(z_i)}$ is identically zero for every treated unit, since every in-sample residual is exactly zero by construction — *not* because the model has actually learned the true $\mu_1(z)$, but because it has simply memorized the training outcomes. AIPW would then behave as though the outcome model were perfectly correct (Exercise 10.2's logic: zero residuals mean the correction contributes nothing, exactly as when $\delta_1 \equiv 0$ in the appendix's proof), when in fact the fitted $\hat{\mu}_1$ may generalize terribly to any new unit. This is precisely why the double/debiased machine learning literature (cited in the appendix) insists on **cross-fitting**: fit $\hat{\mu}_1, \hat{\mu}_0, \hat{e}$ on one part of the data (or a held-out fold) and evaluate the correction terms on a *different* part, so that an overfit nuisance model cannot fool the estimator into reporting artificially small (and misleadingly reassuring) correction terms.

Appendix: Proof Sketch of Double Robustness (Theorem 36.4)

Write $\hat{\mu}_1(z) = \mu_1(z) + \delta_1(z)$ where μ_1 is the true outcome function and δ_1 is the model's error; similarly $\hat{e}(z) = e(z) + \eta(z)$. The AIPW estimand's expectation (for the treated-arm contribution, by symmetry) is

$$\mathbb{E}\left[\hat{\mu}_1(Z) + \frac{T(Y - \hat{\mu}_1(Z))}{\hat{e}(Z)}\right] = \mathbb{E}\left[\mu_1(Z) + \delta_1(Z)\Big(1 - \frac{e(Z)}{\hat{e}(Z)}\Big)\right],$$

using $\mathbb{E}[T \mid Z] = e(Z)$ and $\mathbb{E}[Y \mid T{=}1, Z] = \mu_1(Z)$ (backdoor unconfoundedness). If $\delta_1(Z) \equiv 0$ (outcome model correct), the bias term vanishes regardless of η. If instead $\eta(Z) \equiv 0$ (propensity model correct, so $\hat{e} = e$), the factor $\big(1 - \frac{e(Z)}{\hat{e}(Z)}\big) = \big(1 - \frac{e(Z)}{e(Z)}\big) = 0$ regardless of δ_1. Either condition alone kills the bias term, leaving $\mathbb{E}[\mu_1(Z)] = \mathbb{E}[Y \mid \text{do}(X{=}1)]$ by the backdoor formula — proving consistency under either (but not necessarily both) correct specifications. The full proof, with the analogous control-arm term and a formal treatment of the estimator's asymptotic variance (which is minimized when *both* models are correctly specified, an efficiency bonus beyond mere robustness), is standard in the semiparametric efficiency literature (Robins, Rotnitzky & Zhao, 1994; Chernozhukov et al., 2018, on "double/debiased machine learning," which extends exactly this idea to allow $\hat{\mu}$ and $\hat{e}$ to be arbitrary flexible ML models via sample-splitting).

Connections and Takeaways

This course's arc is now complete. Sessions 1–6 built a language for representing and computing with joint distributions — conditional independence, graphs, exact and approximate inference, and learning. Session 7 showed, with real numbers, that this language cannot by itself answer questions about intervention. Session 8 gave arrows a stronger, mechanistic meaning (SCMs) that supports exactly those questions. Session 9 showed when and how a causal effect can be recovered from observational data and a graph alone (identification). And today closes the loop back to practice: once identification tells us *what* to estimate, the machine learning tools students already know — regression, classification, cross-fitting — tell us *how well* we can estimate it, and the special case of a randomized experiment shows that everything built since Session 7 was, all along, a way of solving — analytically, from observational data — the very problem that random assignment solves for free by design.

Course-level takeaway: causal inference is not a separate discipline bolted onto machine learning. It is machine learning aimed, by a graph-theoretic argument, at a precisely defined and provably estimable target — and knowing when that target is estimable at all, which this course's entire second half was devoted to establishing, is exactly the judgment a working data scientist needs before ever fitting a single model.